USRP X310 Data Recording Report

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Abstract

This report describes the data recording process using the USRP X310 device. This reciver is placed in a fixed position in São José dos Campos, SP, Brazil, in the GPS/GNSS Laboratory at Aeronautics Institute of Technology (ITA)[1]. The present report is updated as new recordings are made and new information is available.



Figure 1: Location of the GNSS antenna.

[1]https://github.com/ita-gnss-lab

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Part I
Recording overview

This part introduces an overview os the recordings presented in this report. It includes the methodology used to record the data, the calendar of the recorded days, the configuration of the USRP X310, and the processing of the recordings with GNSS-SDR.

1 Methodology

The recording is computed automatically by a Linux service which is driven by the following guidelines:

- Everyday, the program will start within the time rage [4PM, 6PM]
- It will sense whether the USRP is detectable via `uhd_find_devices --args=''addr=192.168.10.2''`.
- If not found, it will exit silently and will sense again in intervals of 15 min.
- Once found, two recordings are made, each with duration of 5 minutes (300 s). The first recording happens right after the USRP is detected, and the second recording is made 30 min after the first one. After that, no more recordings will be made for that day.
- The two recordings of a day use different signal pairs:
  - First recording: channel 0 (RX0) GPS L1 C/A (1575.42 MHz), 4 MS/s; channel 1 (RX1) GPS L5 (1176.45 MHz), 20 MS/s; both 25 dB of gain, signed 8-bit I/Q (2 bytes per complex sample);
  - Second recording: channel 0 (RX0) GPS L1 C/A (1575.42 MHz); channel 1 (RX1) GPS L2C (1227.60 MHz); both 4 MS/s, 25 dB of gain, signed 16-bit I/Q (4 bytes per complex sample).
- For each recording a SigMF archive file is created, with the interleaved I/Q samples + metadata, like number of channels (dual channel recording), frequencies, sampling rate, gain, sample type, etc. Each channel has its own configuration, so the metadata of each channel is reported separately.
- Each recording is processed by the GNSS-SDR software receiver right after it is made, and the outputs of every receiver stage are added under the recording in Section II (see Section 3). Galileo E1 shares GPS L1's 1575.42 MHz and Galileo E5a GPS L5's 1176.45 MHz, so the recordings also hold Galileo: the first recording is processed twice (GPS L1 C/A + L5, then Galileo E1 + E5a), and the second once, with GPS L1 C/A, L2C and Galileo E1 together.
- After 15 days of records from the last report. Rubem will send an automatic message to header of the Laboratory, Prof. Dr. Felix Antreich, via WhatsApp with the summarized information from all recorded metadata;

2 Recording calendar

Table 1 shows the days when recordings were made. The color of each day indicates how many recordings were made on that day: green means 2 recordings, blue means 1 recording, and no color means no recording was made.

October

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Table 1: Recording calendar.

3 Validating the recordings with GNSS-SDR

A recording is only useful if a GNSS receiver can use it: find the satellites in it, track their signals, decode their navigation messages, and compute a position. To show that, every recording is processed with GNSS-SDR[2], an open-source software-defined GNSS receiver, and the output of each of its stages is added to the report of that recording (Section II). Figure 2 shows the data chain of GNSS-SDR. Each channel (one per satellite signal) runs its own acquisition, tracking and telemetry decoder; observables and PVT combine all of them.

[2] https://gnss-sdr.org, v0.0.20.

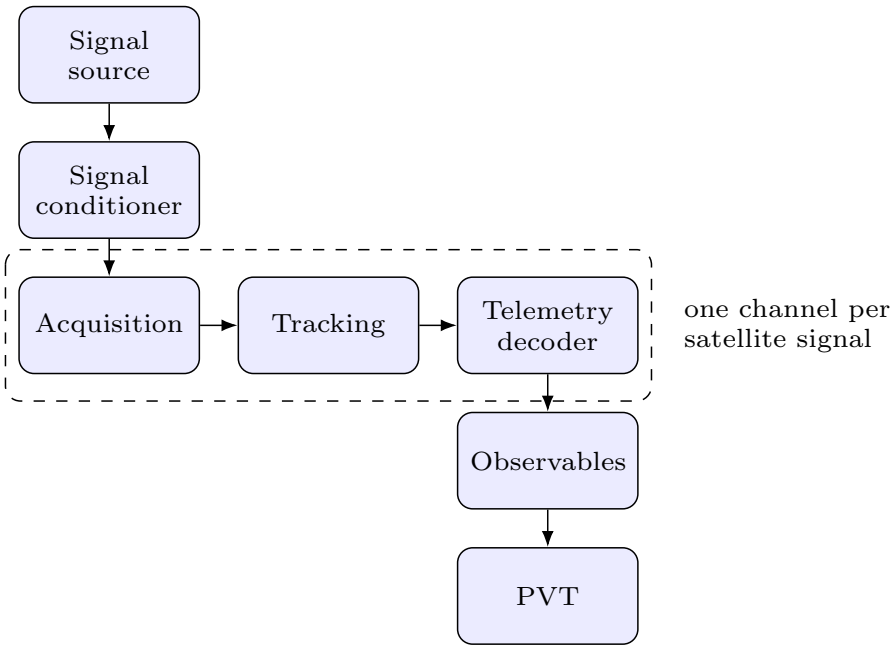


Figure 2: GNSS-SDR receiver chain.

Processing setup. Both channels of a recording are read by one `Multichannel_File_Signal_Source`, so the two bands advance in lock-step (with one independent source per band, one band runs ahead of the other and its measurements are discarded). That source reads both files at one rate: the 4 MS/s L1 C/A file of the L1/L5 recordings is upsampled 5 times (each sample repeated) to the 20 MS/s of L5 before GNSS-SDR. GNSS-SDR v0.0.20's position (RTKLIB, single point) is computed per run as follows:

- **First recording, GPS run** (GPS L1 C/A + L5): from L1 C/A only (broadcast ionosphere model), since its dual-frequency solver was unstable with the few L5-capable GPS satellites in view. The ionosphere-free GPS L1 C/A + L5 position is computed outside GNSS-SDR, from its observables and decoded ephemeris, with our own single-point positioning (`gnss_spp.py`).
- **First recording, Galileo run** (Galileo E1-B/C + E5a): GNSS-SDR's own dual-frequency, ionosphere-free (`Iono-Free-LC`) position. Every Galileo satellite broadcasts both E1 and E5a, so all satellites in view count, unlike GPS L5.
- **Second recording** (GPS L1 C/A + L2C + Galileo E1-B/C): GNSS-SDR's own single-frequency, multi-constellation position from GPS L1 C/A and Galileo E1 together (broadcast ionosphere model; RTKLIB estimates the GPS–Galileo clock offset). GNSS-SDR's solver ignores L2C, so the ionosphere-free GPS L1 C/A + L2C position again comes from `gnss_spp.py`.

The Galileo run's stages are reported after the GPS run's, with "(Galileo E1 + E5a)" in their headings; its signal source is the same recording, so that stage is not repeated.

3.1 Signal source

Reads the recorded I/Q samples (here, the SigMF data files) and delivers them as a stream, one per channel. GNSS-SDR has no output of its own here, so the recording itself is examined.

- **Power spectral density** per channel: the shape of the receiver passband over a flat thermal-noise floor. GNSS signals are 15–20 dB below the noise, so they are invisible here; a narrow peak would be interference.
- **I/Q histograms**: should be Gaussian, centred on zero, and far from the integer limits (no clipping).
- **Table**: format, rate and gain per channel, I/Q standard deviation and mean, percentage of clipped values, and samples lost during the recording (overflows, zero-filled).

3.2 Signal conditioner

Adapts the samples to what the channels process: converts the file format (signed 8- or 16-bit interleaved I/Q) to complex floating point (data type adapter), optionally filters (input filter) and resamples (resampler).

- **Table** per channel: file format, data type adapter, filter, resampling, and rate before and after.

3.3 Acquisition

Detects which satellites are present and gives a first, coarse estimate of each one's code delay and Doppler shift, by correlating the signal with the satellite's spreading code over a two-dimensional grid of delays and Doppler frequencies, and comparing the peak with a detection threshold (set by the probability of false alarm). With dual-frequency assistance, the second band (GPS L5 or L2C, Galileo E5a) is only searched for satellites already tracked on L1/E1, around their L1/E1 Doppler scaled to the second frequency. Galileo E1 uses a 4-ms code (BOC(1,1) modulation), so its correlation peak is narrower and has side peaks.

- **Delay–Doppler map** (surface plot) of the strongest detection per band: a single sharp peak well above the floor.
- **Table**: search attempts, detections and detected satellites, and the best test statistic relative to the threshold. Attempts on satellites that are not in view, or that do not broadcast that band (L5 is only on Block IIF and III satellites, L2C on IIR-M, IIF and III), are rejected.

3.4 Tracking

Follows each detected signal with a delay-lock loop (code) and a phase-lock loop (carrier), producing every 1–20 ms the prompt correlator output (which carries the navigation data bits), the carrier-to-noise density ratio C/N_0, the Doppler, the code, and carrier phases.

- C/N_0 **and Doppler versus time** per satellite and band: C/N_0 of roughly 35–50 dB-Hz for an open-sky antenna, and smooth Doppler curves following the satellite motion.
- **Prompt I/Q scatter** of the strongest satellite per band: with the carrier phase locked, the energy sits on the in-phase axis in two clusters (the ±1 data bits), or in for the dataless pilot components that Galileo is tracked on (E1-C, E5a-Q), the ±1 chips of their secondary code; without phase lock it would be a ring. Each point is one prompt correlator output, the received samples $x[n]$ multiplied by the local code replica c aligned at the estimated code delay $\hat{\tau}$ and by the local carrier replica (estimated Doppler $\hat{f}_D$ and phase $\hat{\phi}$), summed over the N samples of one integration period (T_s: sampling period):

$$P = \sum_{n=1}^{N} x[n]\, c[n - \hat{\tau}]\, e^{-j(2\pi \hat{f}_D n T_s + \hat{\phi})} = I + jQ. \tag{1}$$

- **Loop errors versus time** per satellite and band: the PLL carrier-phase error (degrees) and the DLL code-delay error (chips), i.e. the discriminator outputs the loops drive to zero. Noise centred on zero means clean tracking; jumps would mark cycle slips or a loop losing lock.
- **Table** per satellite: median C/N_0, percentage of time phase-locked, losses of lock, time tracked, and the standard deviation of the PLL and DLL errors.

3.5 Telemetry decoder

Synchronizes to the navigation-message bits and frames, checks their parity/CRC (Cyclic Redundancy Check), decodes the ephemeris (orbit and satellite clock), and time-stamps every measurement with the satellite's transmit time (time of week). GPS L1 C/A carries the legacy LNAV message (6-s subframes); L2C and L5 carry CNAV (12-s and 6-s messages). Galileo E1-B carries I/NAV (2-s pages) and E5a-I F/NAV (10-s pages).

- **Timeline** per band and satellite of when a valid transmit time was available.
- **Table**: messages decoded, ephemeris obtained, and when the first valid time of week came.

3.6 Observables

Aligns all channels on common receiver epochs and forms, for each satellite and band, the pseudorange (from transmit and receive times), the

accumulated carrier phase and the Doppler. These are the measurements a positioning algorithm uses; GNSS-SDR also writes them to a RINEX observation file.

- **Availability** of valid pseudoranges per band and satellite over the recording.
- **Second band minus L1/E1 pseudorange** per satellite: the same constant for every satellite (the receiver's inter-band hardware bias) plus the slowly varying, satellite-dependent ionospheric delay difference (metres). It proves that both bands measure the same satellites consistently.
- **Table** per satellite: common epochs, mean, standard deviation and drift of that difference.

3.7 PVT

Computes position, velocity and receiver clock (PVT) from the pseudoranges and the decoded ephemeris. The antenna position was surveyed: −23.209433°, −45.874875°, 596.42 m (WGS-84), so the errors below are true errors, not just scatter.

- **Horizontal error scatter and east/north/up error versus time** for GNSS-SDR's own solution of each run and, for the GPS runs, `gnss_spp.py`'s dual-frequency ionosphere-free solution. Single-frequency single-point positioning is expected to stay within a few metres horizontally and ~10 m vertically; the iono-free combination removes the ionospheric bias but amplifies the code noise about three times.
- **Sky plot** of the GPS satellites used by `gnss_spp.py`.
- **Table**: mean error (accuracy), standard deviation (precision), 2-D and 3-D RMS, and 95% horizontal error.

Part II
Recording reports

4 Records on 08/10/2026

4.1 First record

Parameter	**Channel 0**	**Channel 1**
Start time	2026-10-08 22:26:37 UTC	
Frequency	1575.420 MHz (L1)	1176.450 MHz (L5)
Duration	300.0 s	
Sample rate	2.000 MS/s	
Sample type	signed complex 16-bit samples (I+Q), little-endian	
Gain	20.0 dB	20.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	2400.0 MB
Samples written	600,000,000	600,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	

Table 2: Metadata for recording `x310_20261008T222630Z`, captured starting 2026-10-08T22:26:37.215640Z.

5 Records on 13/10/2026

5.1 First record

Parameter	**Channel 0**	**Channel 1**
Start time	2026-10-13 21:57:56 UTC	
Frequency	1575.420 MHz (L1)	1176.450 MHz (L5)
Duration	300.0 s	
Sample rate	2.000 MS/s	
Sample type	signed complex 16-bit samples (I+Q), little-endian	
Gain	20.0 dB	20.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	2400.0 MB
Samples written	600,000,000	600,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	

Table 3: Metadata for recording `x310_20261013T215749Z`, captured starting 2026-10-13T21:57:56.746726Z.

5.2 Second record

Parameter	**Channel 0**	**Channel 1**
Start time	2026-10-13 22:08:41 UTC	
Frequency	1575.420 MHz (L1)	1176.450 MHz (L5)
Duration	300.0 s	
Sample rate	2.000 MS/s	
Sample type	signed complex 16-bit samples (I+Q), little-endian	
Gain	20.0 dB	20.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	2400.0 MB
Samples written	600,000,000	600,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	

Table 4: Metadata for recording `x310_20261013T220835Z`, captured starting 2026-10-13T22:08:41.589855Z.

6 Records on 22/10/2026

6.1 First record

Parameter	Channel 0	Channel 1
Start time	2026-10-22 19:00:06 UTC	
Frequency	1575.420 MHz (L1)	1176.450 MHz (L5)
Duration	300.0 s	300.0 s
Sample rate	4.000 MS/s	20.000 MS/s
Sample type	signed complex 8-bit samples (I+Q), converted on the host from the X310's sc16	
Gain	25.0 dB	25.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	12000.0 MB
Samples written	1,200,000,000	6,000,000,000
Lost samples	2912	14924
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	
GNSS-SDR configuration	`l1ca8bits4msps_l5_8bits20msps.conf`, `galileo_e1_8bits4msps_e5a_8bits20msps.conf`	

Table 5: Metadata for recording `x310_20261022T190000Z`, captured starting 2026-10-22T19:00:06.973177Z.

6.1.1 Signal source

Raw I/Q samples of each channel analysed within the window [10, 10.5] s after the start recording instant.

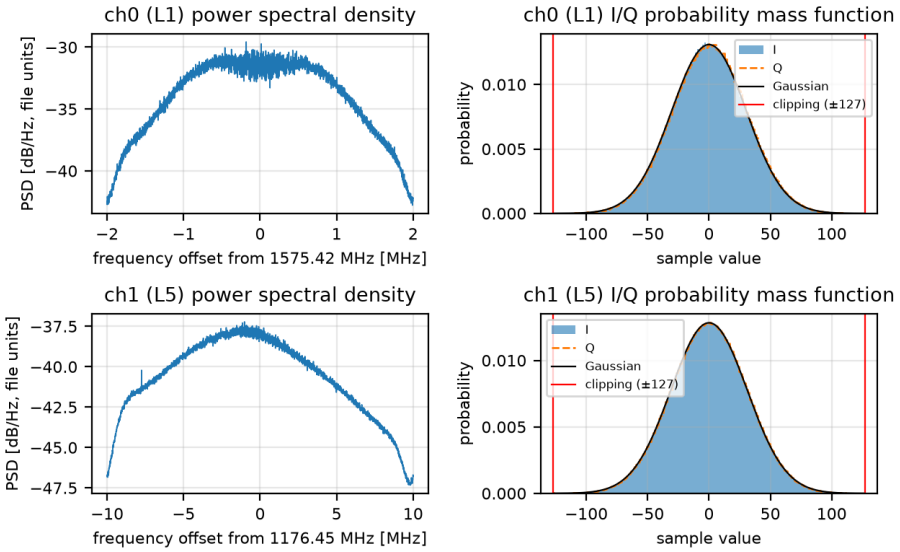


Figure 3: Signal source: power spectral density (the receiver passband over a flat thermal-noise floor; GNSS signals are below the noise, so only interference would stand out) and probability mass functions of the integer I/Q sample values (Gaussian noise, centred, not clipped).

Channel	**ch0**	**ch1**
Frequency [MHz]	1575.42	1176.45
Rate [MS/s]	4	20
Format	ci8	ci8
Bytes per complex sample	2 (1 for I, 1 for Q)	2 (1 for I, 1 for Q)
Full scale	±127	±127
Gain [dB]	25	25
Std I / Q	30.4 / 30.5	31.1 / 31.1
Mean I / Q	-0.26 / 0.20	0.26 / -0.07
Min / max I	-127 / 127	-127 / 127
Min / max Q	-127 / 127	-127 / 127
Headroom[3]	4.2	4.1
Clipped [%]	0.003	0.004
Lost samples	2,912	14,924

Table 6: Signal source: sample statistics per channel (headroom: how many standard deviations of the noise fit before clipping).

[3]Headroom = A/σ, where A is the full scale of the sample format (127 for 8-bit, 32767 for 16-bit) and σ the standard deviation of the I and Q samples together (both in file units, over the analysed window). For Gaussian noise, the fraction of clipped values is $2Q(A/\sigma)$, with Q the Gaussian tail function: e.g. 0.003 % at 4.2σ.

6.1.2 Signal conditioner

Channel	ch0 (L1 C/A)	ch1 (L5)
File format	ci8	ci8
Source item type	ibyte	ibyte
Data type adapter	Ibyte_To_Complex	Ibyte_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	×5 zero-order hold, offline (`upsample_iq.py`)	Pass_Through
Rate [MS/s]	4 → 20	20 → 20

Table 7: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

6.1.3 Acquisition

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

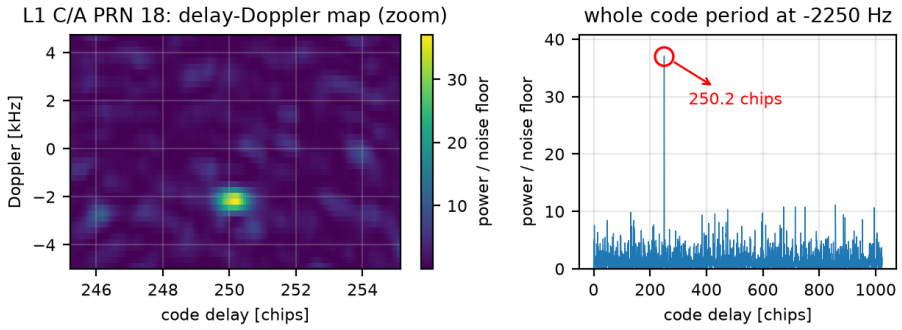


Figure 4: Acquisition, L1 C/A: delay–Doppler search of PRN 18; one sharp peak (test statistic 50 vs threshold 47) at -2250 Hz and 250.2 chips.

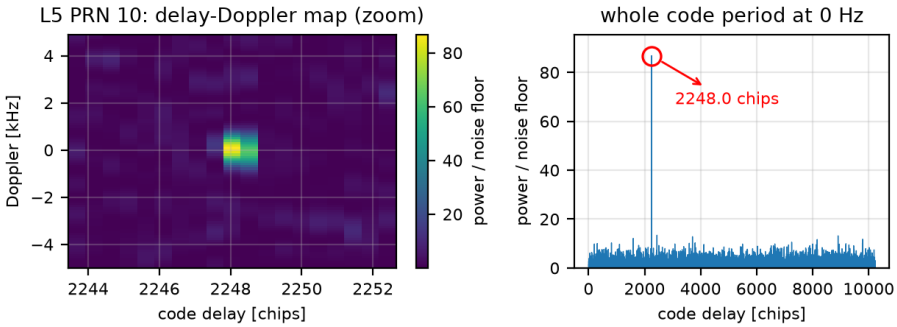


Figure 5: Acquisition, L5: delay–Doppler search of PRN 10; one sharp peak (test statistic 122 vs threshold 49) at -1491 Hz and 2248.0 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold[4]
L1 C/A	9	1	18	1.1
L5	1	1	10	2.5

Table 8: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

[4]Margin of the band's strongest detection: the largest acquisition test statistic among the detected attempts (the correlation peak of the delay–Doppler grid, normalised by the power of the input samples searched, i.e. the noise power, since GNSS signals are below the noise) divided by the detection threshold, which GNSS-SDR derives from the configured probability of false alarm. A value of 1 is right at the detection limit; e.g. 2.5 means the peak cleared the threshold 2.5 times over.

6.1.4 Tracking

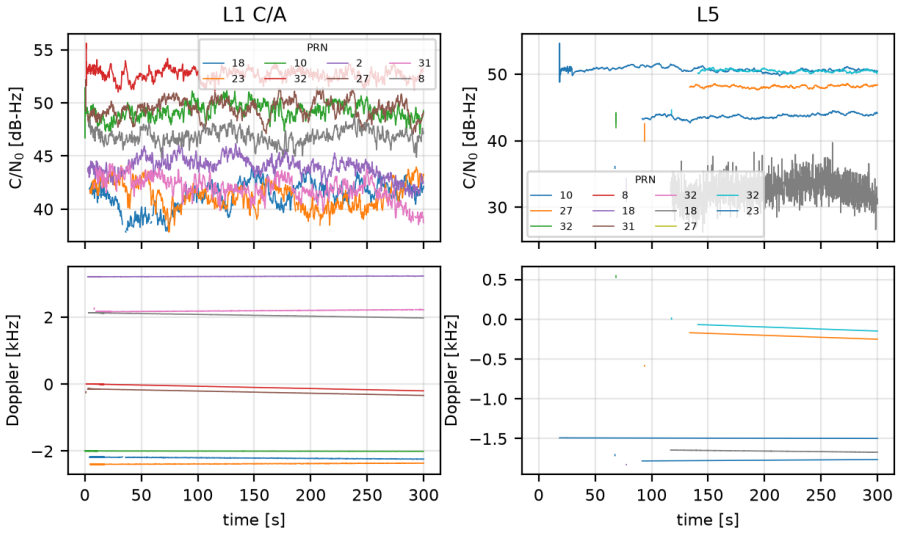


Figure 6: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

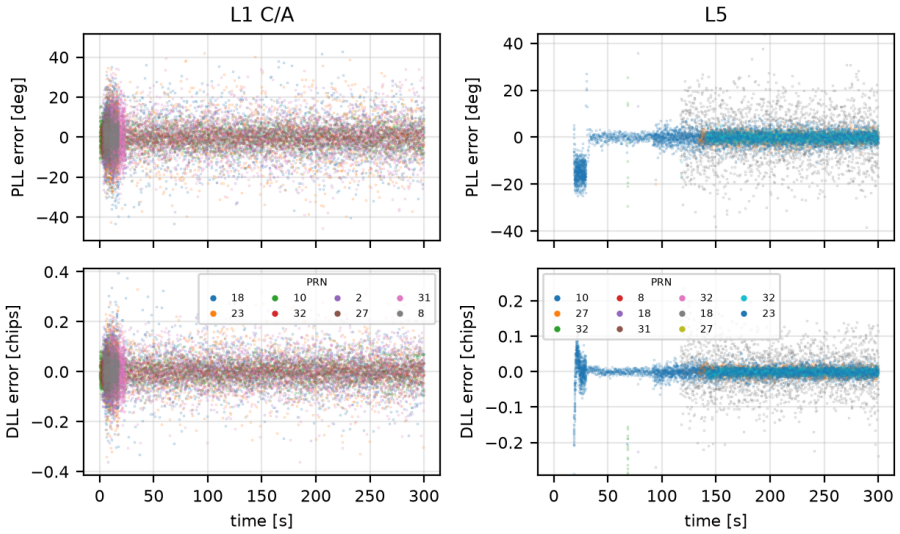


Figure 7: Tracking: loop errors per satellite while phase-locked, i.e. the discriminator outputs that the loops drive to zero: PLL carrier-phase error (degrees) and DLL code-delay error (chips) of every correlation. Noise centred on zero = clean tracking; its spread grows as C/N_0 drops, and jumps would mark cycle slips or a loop losing lock.

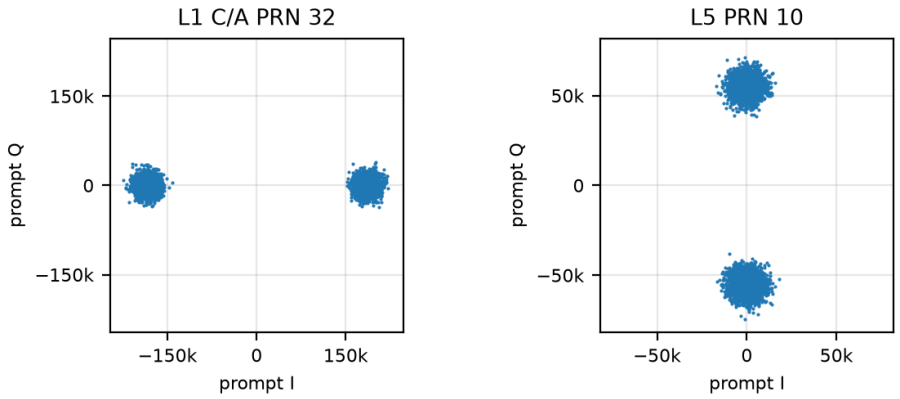


Figure 8: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or, for the dataless pilots tracked for Galileo (E1-C, E5a-Q), the ±1 chips of their secondary code, mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]	PLL error std [deg]	DLL error std [chips]
L1 C/A	0	18	41.7	93	0	298	12.1	0.090
L1 C/A	1	23	41.9	93	0	297	12.1	0.088
L1 C/A	2	10	49.2	100	0	300	5.0	0.037
L1 C/A	3	32	52.9	100	0	299	3.4	0.025
L1 C/A	4	2	43.9	95	0	299	9.1	0.067
L1 C/A	5	27	49.1	94	0	299	5.7	0.035
L1 C/A	6	31	42.7	94	0	292	11.0	0.080
L1 C/A	7	8	47.0	94	0	298	6.5	0.049
L5	8	10	50.7	100	0	282	7.2	0.028
L5	10	27	48.0	37	2	206	2.5	0.018
L5	10	32	29.0	0	2	16	52.4	0.071
L5	11	8	28.4	0	2	16	–	–
L5	11	18	33.5	0	2	16	27.8	0.200
L5	12	31	29.0	0	2	16	–	–
L5	12	32	28.7	0	2	16	–	–
L5	13	18	32.8	60	1	206	10.8	0.056
L5	13	27	28.8	0	1	16	–	–
L5	14	32	50.3	36	1	198	2.4	0.015
L5	15	23	43.7	63	0	232	3.3	0.019

Table 9: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8; loop error standard deviations while phase-locked).

6.1.5 Telemetry decoder

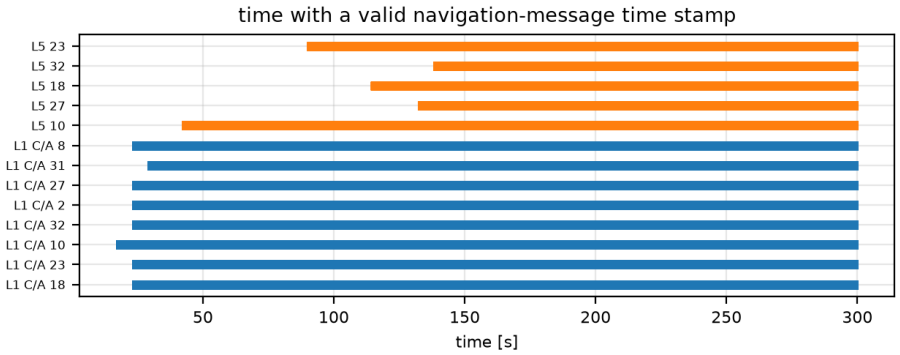


Figure 9: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
L1 C/A	2	47	yes	23.1	414048.0
L1 C/A	8	45	yes	23.1	414048.0
L1 C/A	10	43	yes	17.1	414042.0
L1 C/A	18	47	yes	23.1	414048.0
L1 C/A	23	47	yes	23.1	414048.0
L1 C/A	27	47	yes	23.1	414048.0
L1 C/A	31	46	yes	29.1	414054.0
L1 C/A	32	47	yes	23.1	414048.0
L5	10	26	yes	42.2	414067.1
L5	18	24	yes	114.4	414139.3
L5	23	27	yes	90.0	414114.9
L5	27	17	yes	132.3	414157.2
L5	32	16	yes	138.2	414163.1

Table 10: Telemetry decoder: L1 C/A LNAV subframes, L5 CNAV messages decoded, ephemeris obtained, and the first valid time of week.

6.1.6 Observables

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

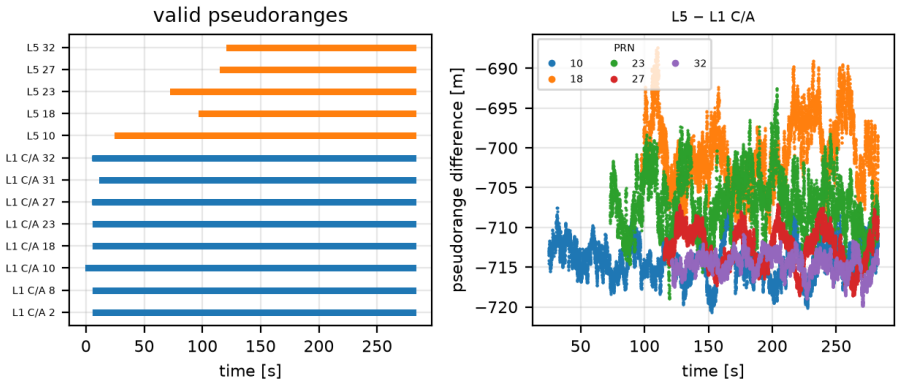


Figure 10: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
L5 − L1 C/A	10	12831	86	-714.11	2.26	0.0005
L5 − L1 C/A	18	9267	62	-700.79	4.40	0.0122
L5 − L1 C/A	23	10483	70	-706.92	3.69	-0.0064
L5 − L1 C/A	27	8373	56	-712.59	2.40	-0.0092
L5 − L1 C/A	32	8078	54	-714.63	1.36	-0.0046

Table 11: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

6.1.7 PVT

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

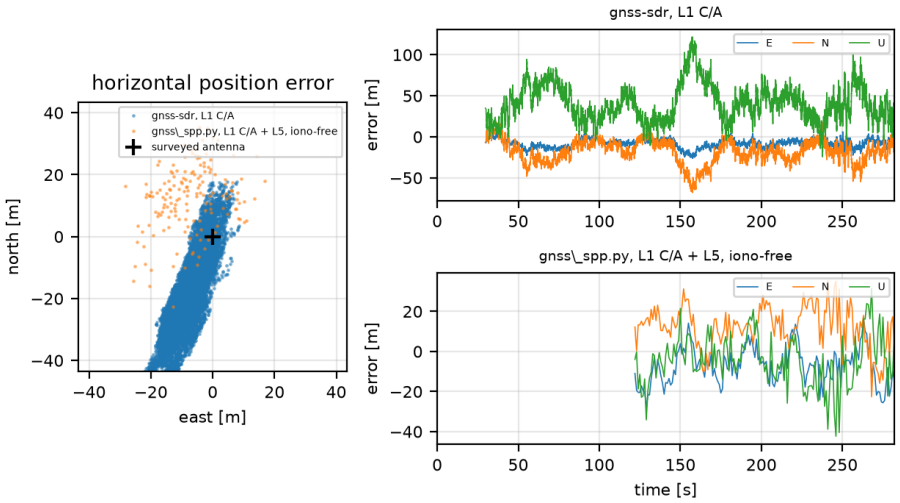


Figure 11: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution (single frequency, broadcast ionosphere model). gnss_spp.py: computed from gnss-sdr's observables and decoded GPS ephemeris, removing the ionospheric delay with the second band (pseudoranges carrier-smoothed over 100 s).

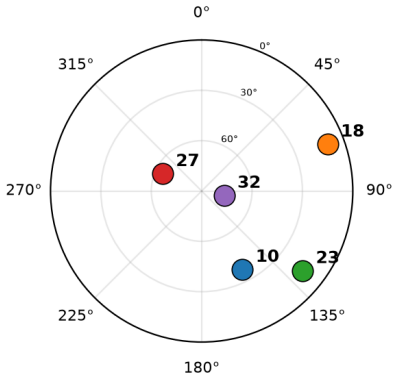


Figure 12: PVT: sky plot of the GPS satellites in the gnss_spp.py dual-frequency solution (azimuth, elevation) during the recording.

Solution	gnss-sdr, L1 C/A	gnss_spp.py, L1 C/A + L5, iono-free
Solutions	13135	161
Run with fix [%]	90	57
Mean E / N / U [m]	-7.6 / -18.0 / 40.8	-7.3 / 12.2 / -4.6
Std E / N / U [m]	5.5 / 14.2 / 22.4	9.3 / 10.4 / 12.8
RMS 2-D [m]	24.8	19.9
RMS 3-D [m]	52.8	24.1
95% horiz. [m]	46.5	29.6

Table 12: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp.py computes one per second).

6.1.8 Signal conditioner (Galileo E1 + E5a)

Channel	ch0 (E1)	ch1 (E5a)
File format	ci8	ci8
Source item type	ibyte	ibyte
Data type adapter	Ibyte_To_Complex	Ibyte_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	×5 zero-order hold, offline (`upsample_iq.py`)	Pass_Through
Rate [MS/s]	4 → 20	20 → 20

Table 13: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

6.1.9 Acquisition (Galileo E1 + E5a)

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

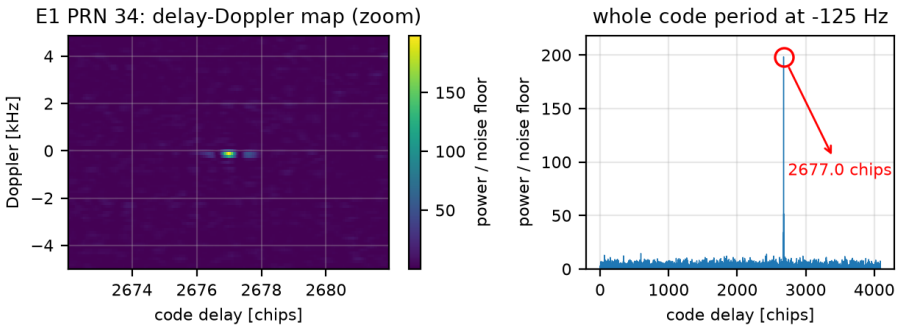


Figure 13: Acquisition, E1: delay–Doppler search of PRN 34; one sharp peak (test statistic 272 vs threshold 52) at -125 Hz and 2677.0 chips.

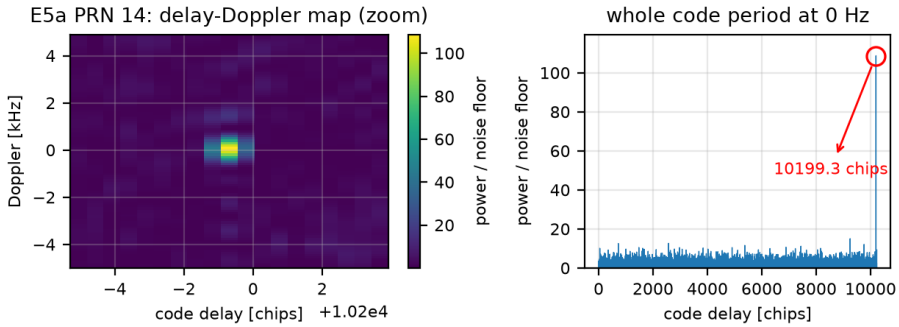


Figure 14: Acquisition, E5a: delay–Doppler search of PRN 14; one sharp peak (test statistic 149 vs threshold 49) at -1427 Hz and 10199.3 chips.

Band	**Attempts**	**Detections**	**Detected PRNs**	**Best statistic / threshold**[5]
E1	5	1	34	5.3
E5a	2	2	14	3.0

Table 14: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

[5]Margin of the band's strongest detection: the largest acquisition test statistic among the detected attempts (the correlation peak of the delay–Doppler grid, normalised by the power of the input samples searched, i.e. the noise power, since GNSS signals are below the noise) divided by the detection threshold, which GNSS-SDR derives from the configured probability of false alarm. A value of 1 is right at the detection limit; e.g. 2.5 means the peak cleared the threshold 2.5 times over.

6.1.10 Tracking (Galileo E1 + E5a)

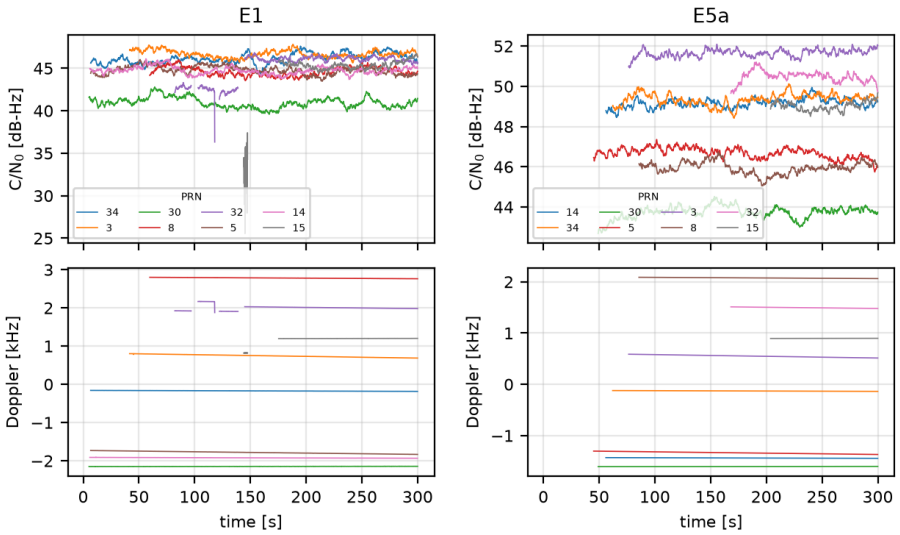


Figure 15: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

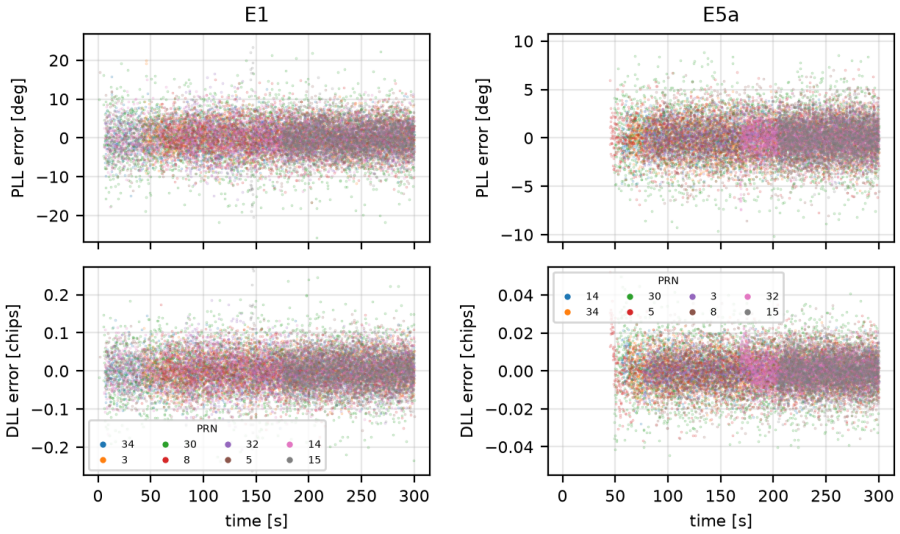


Figure 16: Tracking: loop errors per satellite while phase-locked, i.e. the discriminator outputs that the loops drive to zero: PLL carrier-phase error (degrees) and DLL code-delay error (chips) of every correlation. Noise centred on zero = clean tracking; its spread grows as C/N_0 drops, and jumps would mark cycle slips or a loop losing lock.

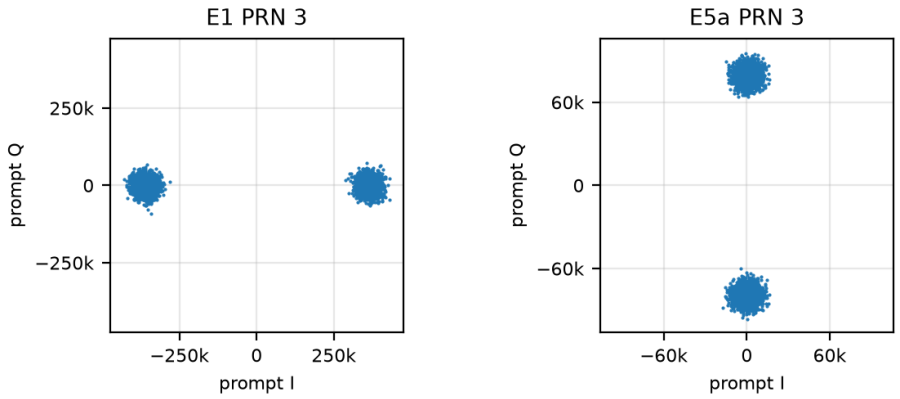


Figure 17: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or, for the dataless pilots tracked for Galileo (E1-C, E5a-Q), the ±1 chips of their secondary code, mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]	PLL error std [deg]	DLL error std [chips]
E1	0	34	46.1	98	0	299	3.7	0.039
E1	1	3	46.6	87	2	298	3.6	0.036
E1	2	30	40.9	99	0	299	6.7	0.071
E1	3	8	44.4	81	2	298	4.3	0.047
E1	4	32	45.7	68	7	299	4.1	0.043
E1	5	5	44.8	98	0	298	4.4	0.046
E1	6	14	44.7	98	0	300	4.0	0.042
E1	7	15	44.5	43	8	300	4.6	0.048
E5a	8	14	48.9	31	1	275	1.6	0.008
E5a	9	34	48.9	30	1	274	1.6	0.008
E5a	10	30	43.5	52	0	272	3.0	0.015
E5a	11	5	46.8	53	0	274	2.2	0.011
E5a	12	3	51.6	49	0	244	1.4	0.006
E5a	13	8	45.9	48	0	235	2.3	0.011
E5a	14	32	50.4	37	0	144	2.0	0.011
E5a	15	15	49.0	30	0	114	1.7	0.008

Table 15: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8; loop error standard deviations while phase-locked).

6.1.11 Telemetry decoder (Galileo E1 + E5a)

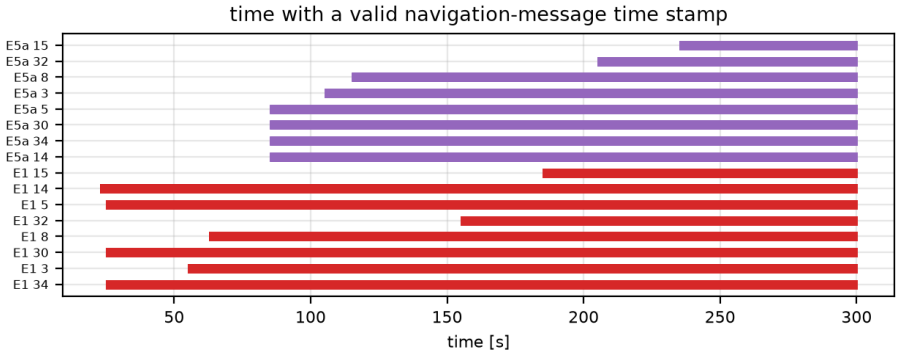


Figure 18: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
E1	3	24	yes	55.2	414080.0
E1	5	26	yes	25.2	414050.0
E1	8	23	yes	63.2	414088.0
E1	14	26	yes	23.2	414048.0
E1	15	10	yes	185.2	414210.0
E1	30	26	yes	25.2	414050.0
E1	32	13	yes	155.2	414180.0
E1	34	26	yes	25.2	414050.0
E5a	3	11	yes	105.4	414130.3
E5a	5	12	yes	85.4	414110.3
E5a	8	10	yes	115.4	414140.3
E5a	14	11	yes	85.4	414110.3
E5a	15	3	yes	235.4	414260.3
E5a	30	12	yes	85.4	414110.3
E5a	32	5	yes	205.4	414230.3
E5a	34	12	yes	85.4	414110.3

Table 16: Telemetry decoder: E1 I/NAV words, E5a F/NAV pages decoded, ephemeris obtained, and the first valid time of week.

6.1.12 Observables (Galileo E1 + E5a)

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

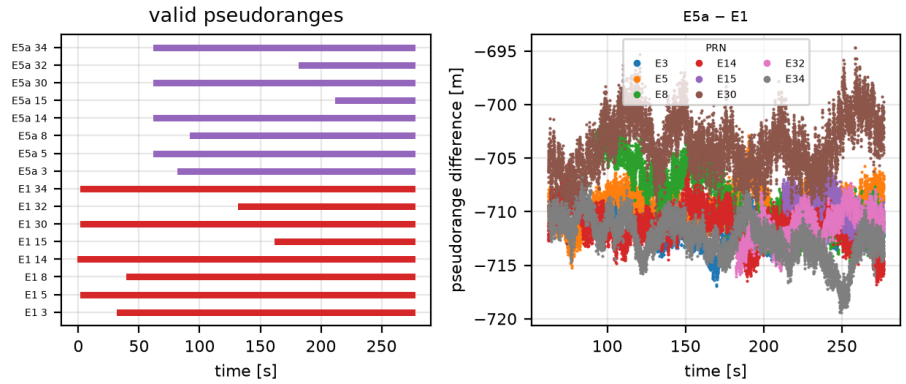


Figure 19: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
E5a − E1	3	9563	64	-711.43	1.45	-0.0075
E5a − E1	5	10437	70	-709.61	1.55	0.0065
E5a − E1	8	9128	61	-708.81	2.89	-0.0451
E5a − E1	14	10437	70	-711.16	1.64	-0.0082
E5a − E1	15	3147	21	-709.62	1.46	-0.0504
E5a − E1	30	10457	70	-703.72	2.59	0.0079
E5a − E1	32	4647	31	-711.39	1.51	0.0347
E5a − E1	34	10437	70	-712.45	1.88	-0.0177

Table 17: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

6.1.13 PVT (Galileo E1 + E5a)

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

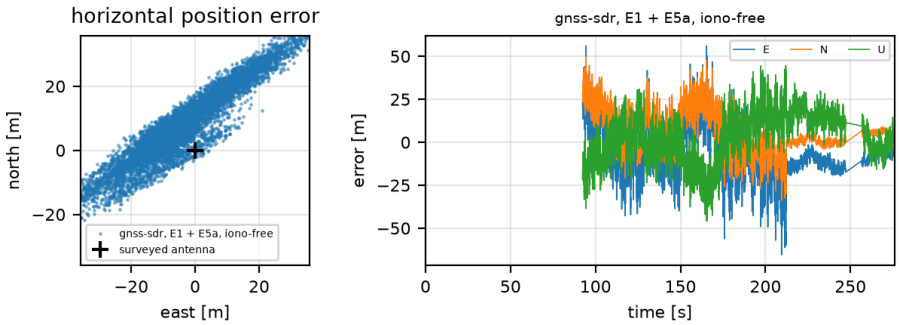


Figure 20: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution.

Solution	**gnss-sdr, E1 + E5a, iono-free**
Solutions	8642
Run with fix [%]	64
Mean E / N / U [m]	-4.8 / 6.7 / 2.8
Std E / N / U [m]	15.5 / 11.9 / 14.2
RMS 2-D [m]	21.2
RMS 3-D [m]	25.6
95% horiz. [m]	40.3

Table 18: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp.py computes one per second).

6.2 Second record

Parameter	Channel 0	Channel 1
Start time	2026-10-22 19:30:07 UTC	
Frequency	1575.420 MHz (L1)	1227.600 MHz (L2)
Duration	300.0 s	300.0 s
Sample rate	4.000 MS/s	4.000 MS/s
Sample type	signed complex 16-bit samples (I+Q), little-endian, as received from the X310 (sc16)	
Gain	25.0 dB	25.0 dB
Antenna	RX1	RX2
File size	4800.0 MB	4800.0 MB
Samples written	1,200,000,000	1,200,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	
GNSS-SDR configuration	`l1ca16bits4msps_l2c16bits4msps.conf`	

Table 19: Metadata for recording `x310_20261022T193000Z`, captured starting 2026-10-22T19:30:07.040026Z.

6.2.1 Signal source

Raw I/Q samples of each channel analysed within the window [10, 10.5] s after the start recording instant.

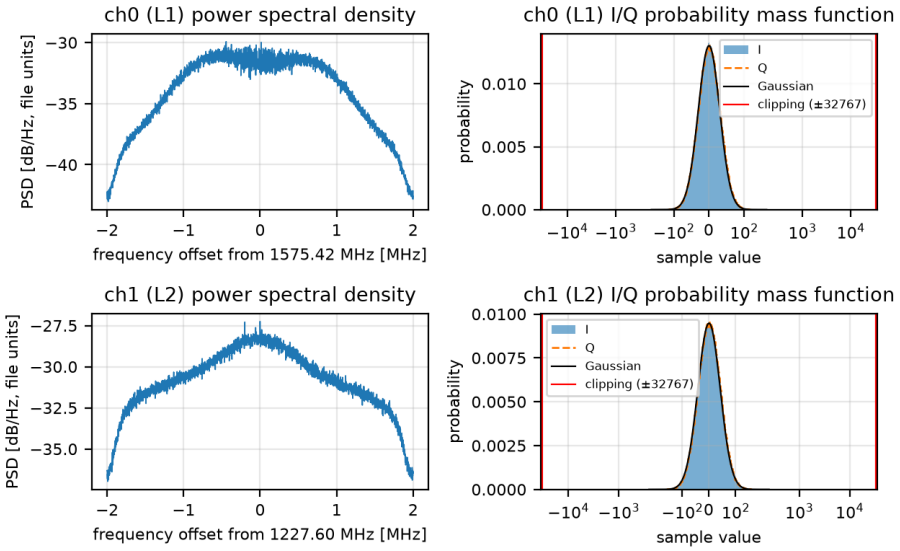


Figure 21: Signal source: power spectral density (the receiver passband over a flat thermal-noise floor; GNSS signals are below the noise, so only interference would stand out) and probability mass functions of the integer I/Q sample values (Gaussian noise, centred, not clipped).

Channel	**ch0**	**ch1**
Frequency [MHz]	1575.42	1227.60
Rate [MS/s]	4	4
Format	ci16_le	ci16_le
Bytes per complex sample	4 (2 for I, 2 for Q)	4 (2 for I, 2 for Q)
Full scale	±32767	±32767
Gain [dB]	25	25
Std I / Q	30.2 / 31.1	42.1 / 42.1
Mean I / Q	-0.04 / -0.13	0.28 / 0.05
Min / max I	-151 / 154	-227 / 208
Min / max Q	-150 / 164	-208 / 210
Headroom[6]	1069.3	778.5
Clipped [%]	0.000	0.000
Lost samples	0	0

Table 20: Signal source: sample statistics per channel (headroom: how many standard deviations of the noise fit before clipping).

[6]Headroom = A/σ, where A is the full scale of the sample format (127 for 8-bit, 32767 for 16-bit) and σ the standard deviation of the I and Q samples together (both in file units, over the analysed window). For Gaussian noise, the fraction of clipped values is $2Q(A/\sigma)$, with Q the Gaussian tail function: e.g. 0.003 % at 4.2σ.

6.2.2 Signal conditioner

Channel	ch0 (L1 C/A, E1)	ch1 (L2C)
File format	ci16_le	ci16_le
Source item type	ishort	ishort
Data type adapter	Ishort_To_Complex	Ishort_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	Pass_Through	Pass_Through
Rate [MS/s]	4 → 4	4 → 4

Table 21: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

6.2.3 Acquisition

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

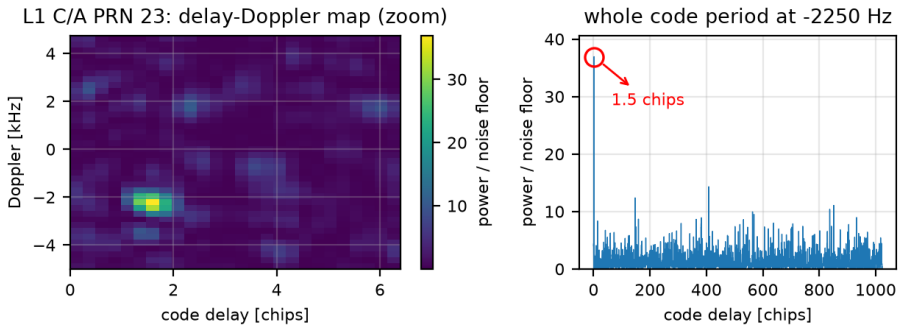


Figure 22: Acquisition, L1 C/A: delay–Doppler search of PRN 23; one sharp peak (test statistic 52 vs threshold 44) at -2250 Hz and 1.5 chips.

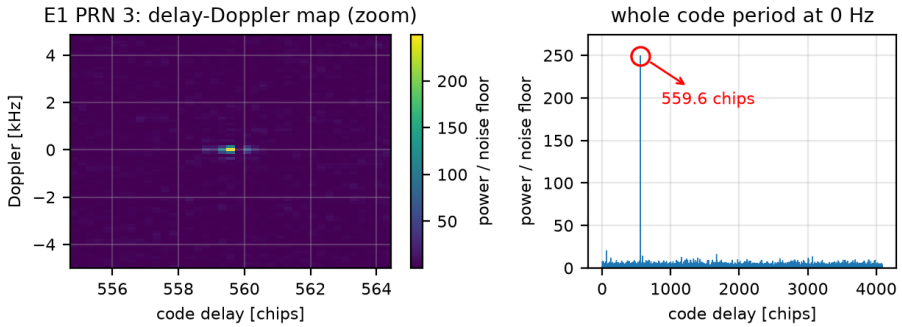


Figure 23: Acquisition, E1: delay–Doppler search of PRN 3; one sharp peak (test statistic 338 vs threshold 48) at 0 Hz and 559.6 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold[7]
L1 C/A	3	1	23	1.2
L2C	not dumped (search grid too large); acquired satellites: see the tracking table			
E1	6	1	3	7.0

Table 22: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

[7]Margin of the band's strongest detection: the largest acquisition test statistic among the detected attempts (the correlation peak of the delay–Doppler grid, normalised by the power of the input samples searched, i.e. the noise power, since GNSS signals are below the noise) divided by the detection threshold, which GNSS-SDR derives from the configured probability of false alarm. A value of 1 is right at the detection limit; e.g. 2.5 means the peak cleared the threshold 2.5 times over.

6.2.4 Tracking

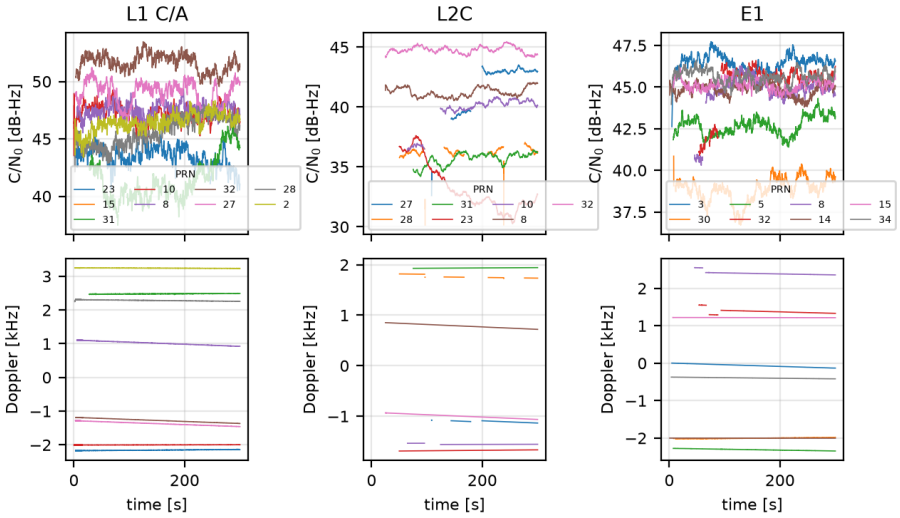


Figure 24: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

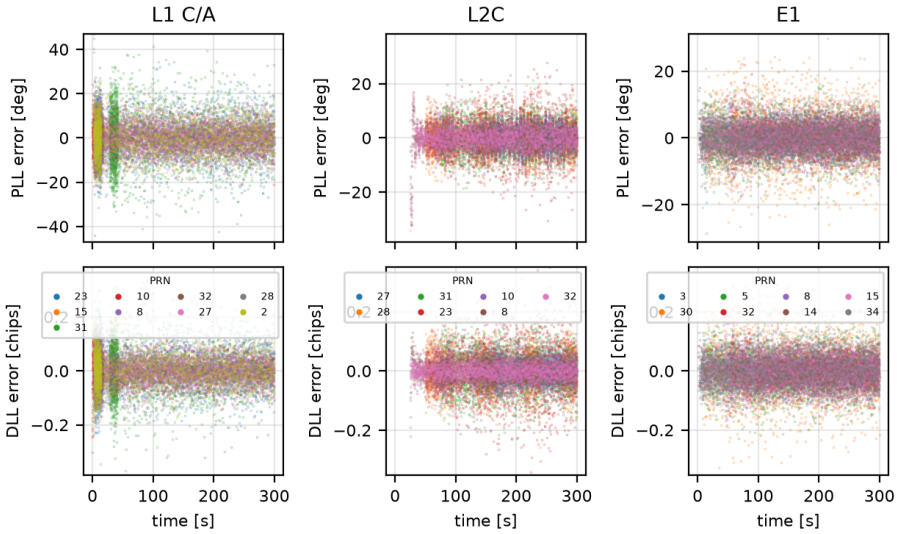


Figure 25: Tracking: loop errors per satellite while phase-locked, i.e. the discriminator outputs that the loops drive to zero: PLL carrier-phase error (degrees) and DLL code-delay error (chips) of every correlation. Noise centred on zero = clean tracking; its spread grows as C/N_0 drops, and jumps would mark cycle slips or a loop losing lock.

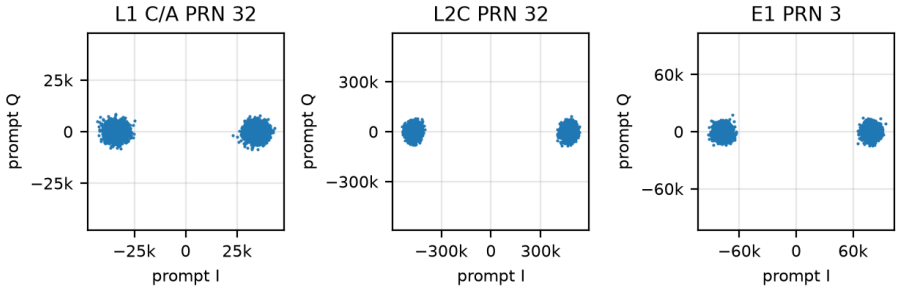


Figure 26: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or, for the dataless pilots tracked for Galileo (E1-C, E5a-Q), the ±1 chips of their secondary code, mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]	PLL error std [deg]	DLL error std [chips]
L1 C/A	0	23	43.6	94	0	298	9.7	0.076
L1 C/A	1	15	28.7	0	1	16	–	–
L1 C/A	1	31	42.2	95	1	273	11.9	0.091
L1 C/A	2	10	46.9	100	0	300	6.4	0.052
L1 C/A	3	8	47.4	92	0	297	6.2	0.050
L1 C/A	4	32	51.4	95	0	298	3.9	0.028
L1 C/A	5	27	48.7	96	0	298	5.2	0.038
L1 C/A	6	28	45.0	96	0	298	8.6	0.059
L1 C/A	7	2	45.8	100	0	298	7.3	0.057
L2C	8	27	39.5	50	3	275	4.6	0.031
L2C	9	28	36.3	49	3	275	6.8	0.058
L2C	10	31	35.8	90	0	251	5.8	0.061
L2C	12	23	32.7	91	0	276	7.7	0.087
L2C	13	10	40.0	75	1	276	4.1	0.039
L2C	14	8	41.1	100	0	275	4.3	0.033
L2C	15	32	44.8	100	0	275	3.7	0.022
E1	16	3	46.5	100	0	297	3.6	0.040
E1	17	30	38.9	99	0	292	8.6	0.096
E1	18	5	42.7	99	0	297	5.5	0.063
E1	19	32	45.5	80	4	293	3.9	0.046
E1	20	8	44.8	84	3	297	4.5	0.050
E1	21	14	44.9	100	0	299	3.8	0.045
E1	22	15	45.1	98	0	299	4.0	0.044
E1	23	34	45.6	100	0	297	3.9	0.044

Table 23: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8; loop error standard deviations while phase-locked).

6.2.5 Telemetry decoder

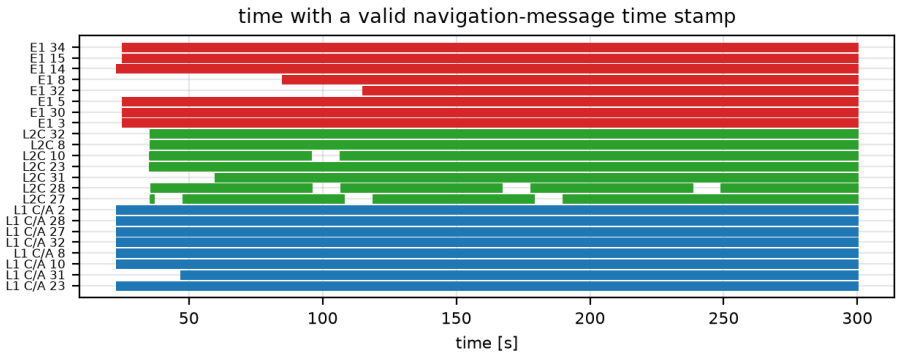


Figure 27: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
E1	3	27	yes	25.1	415850.0
E1	5	26	yes	25.1	415850.0
E1	8	21	yes	85.1	415910.0
E1	14	26	yes	23.1	415848.0
E1	15	27	yes	25.1	415850.0
E1	30	27	yes	25.1	415850.0
E1	32	18	yes	115.1	415940.0
E1	34	27	yes	25.1	415850.0
L1 C/A	2	42	yes	23.0	415848.0
L1 C/A	8	47	yes	23.0	415848.0
L1 C/A	10	43	yes	23.0	415848.0
L1 C/A	23	47	yes	23.0	415848.0
L1 C/A	27	47	yes	23.0	415848.0
L1 C/A	28	47	yes	23.0	415848.0
L1 C/A	31	43	yes	47.0	415872.0
L1 C/A	32	47	yes	23.0	415848.0
L2C	8	13	yes	35.5	0.0
L2C	10	9	yes	35.4	0.0
L2C	23	15	yes	35.4	0.0
L2C	27	5	yes	35.7	0.0
L2C	28	–	no	35.8	0.0
L2C	31	12	yes	59.8	0.0
L2C	32	13	yes	35.6	0.0

Table 24: Telemetry decoder: L1 C/A LNAV subframes, L2C CNAV messages, E1 I/NAV words decoded, ephemeris obtained, and the first valid time of week.

6.2.6 Observables

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

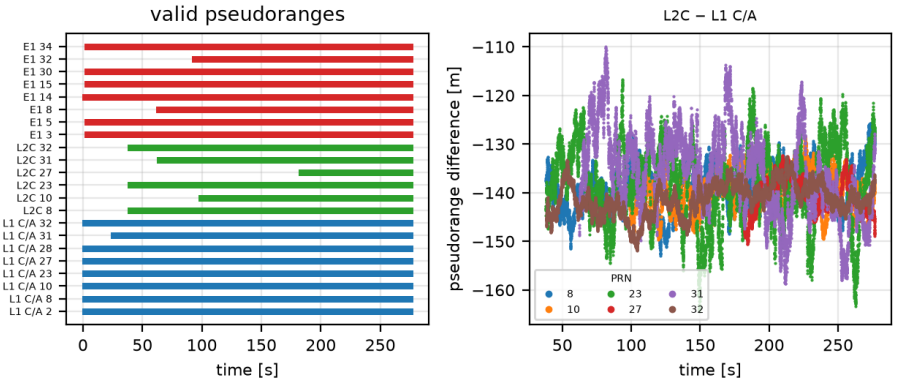


Figure 28: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
L2C – L1 C/A	8	11932	80	-138.80	4.22	0.0218
L2C – L1 C/A	10	8950	60	-139.45	3.99	0.0142
L2C – L1 C/A	23	11937	80	-138.23	7.93	-0.0048
L2C – L1 C/A	27	4756	32	-140.52	3.36	-0.0064
L2C – L1 C/A	31	10717	72	-135.85	8.56	-0.0721
L2C – L1 C/A	32	11928	80	-141.29	3.29	0.0180

Table 25: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

6.2.7 PVT

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

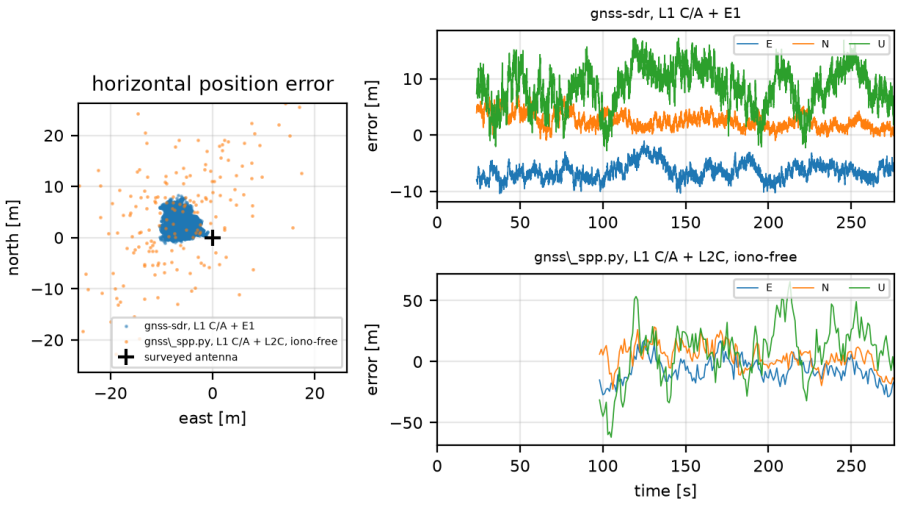


Figure 29: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution (single frequency, broadcast ionosphere model). gnss_spp.py: computed from gnss-sdr's observables and decoded GPS ephemeris, removing the ionospheric delay with the second band (pseudoranges carrier-smoothed over 100 s).

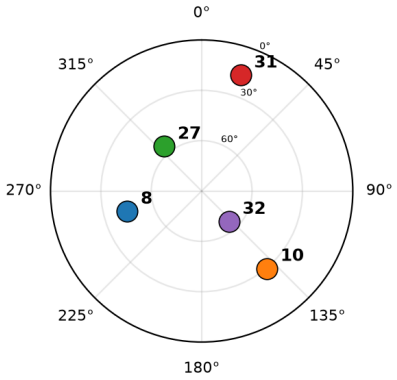


Figure 30: PVT: sky plot of the GPS satellites in the gnss_spp.py dual-frequency solution (azimuth, elevation) during the recording.

Solution	**gnss-sdr, L1 C/A + E1**	**gnss_spp.py, L1 C/A + L2C, iono-free**
Solutions	13139	179
Run with fix [%]	92	65
Mean E / N / U [m]	-6.5 / 2.6 / 8.4	-6.9 / 4.1 / 8.3
Std E / N / U [m]	1.5 / 1.4 / 3.7	9.6 / 10.1 / 23.8
RMS 2-D [m]	7.3	16.1
RMS 3-D [m]	11.7	29.9
95% horiz. [m]	9.5	28.1

Table 26: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp.py computes one per second).

7 Records on 23/10/2026

7.1 First record

Parameter	Channel 0	Channel 1
Start time	2026-10-23 19:00:06 UTC	
Frequency	1575.420 MHz (L1)	1176.450 MHz (L5)
Duration	300.0 s	300.0 s
Sample rate	4.000 MS/s	20.000 MS/s
Sample type	signed complex 8-bit samples (I+Q), converted on the host from the X310's sc16	
Gain	25.0 dB	25.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	12000.0 MB
Samples written	1,200,000,000	6,000,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	
GNSS-SDR configuration	`l1ca8bits4msps_l5_8bits20msps.conf`, `galileo_e1_8bits4msps_e5a_8bits20msps.conf`	

Table 27: Metadata for recording `x310_20261023T190000Z`, captured starting 2026-10-23T19:00:06.932068Z.

7.1.1 Signal source

Raw I/Q samples of each channel analysed within the window [10, 10.5] s after the start recording instant.

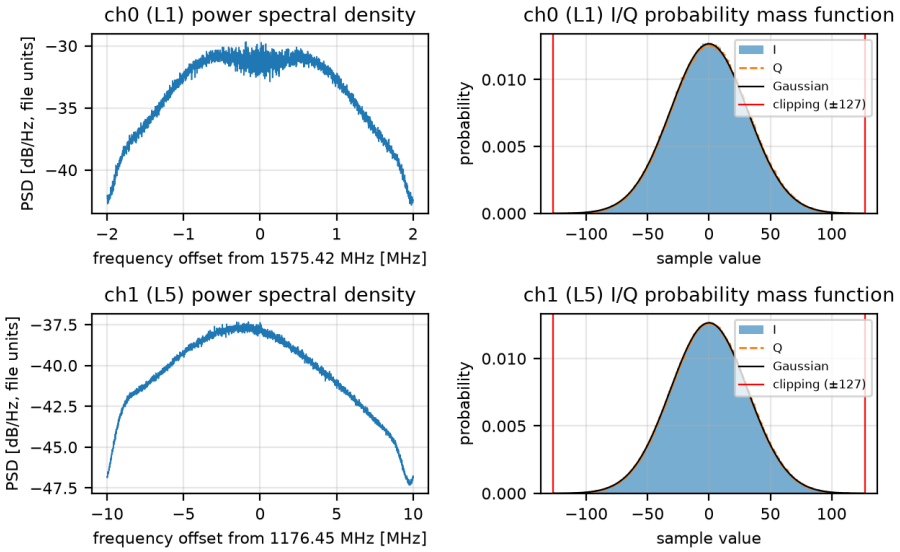


Figure 31: Signal source: power spectral density (the receiver passband over a flat thermal-noise floor; GNSS signals are below the noise, so only interference would stand out) and probability mass functions of the integer I/Q sample values (Gaussian noise, centred, not clipped).

Channel	**ch0**	**ch1**
Frequency [MHz]	1575.42	1176.45
Rate [MS/s]	4	20
Format	ci8	ci8
Bytes per complex sample	2 (1 for I, 1 for Q)	2 (1 for I, 1 for Q)
Full scale	±127	±127
Gain [dB]	25	25
Std I / Q	31.5 / 31.5	31.5 / 31.5
Mean I / Q	0.04 / 0.07	0.04 / 0.01
Min / max I	-127 / 127	-127 / 127
Min / max Q	-127 / 127	-127 / 127
Headroom[8]	4.0	4.0
Clipped [%]	0.006	0.006
Lost samples	0	0

Table 28: Signal source: sample statistics per channel (headroom: how many standard deviations of the noise fit before clipping).

[8]Headroom = A/σ, where A is the full scale of the sample format (127 for 8-bit, 32767 for 16-bit) and σ the standard deviation of the I and Q samples together (both in file units, over the analysed window). For Gaussian noise, the fraction of clipped values is $2Q(A/\sigma)$, with Q the Gaussian tail function: e.g. 0.003 % at 4.2σ.

7.1.2 Signal conditioner

Channel	ch0 (L1 C/A)	ch1 (L5)
File format	ci8	ci8
Source item type	ibyte	ibyte
Data type adapter	Ibyte_To_Complex	Ibyte_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	×5 zero-order hold, offline (`upsample_iq.py`)	Pass_Through
Rate [MS/s]	4 → 20	20 → 20

Table 29: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

7.1.3 Acquisition

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

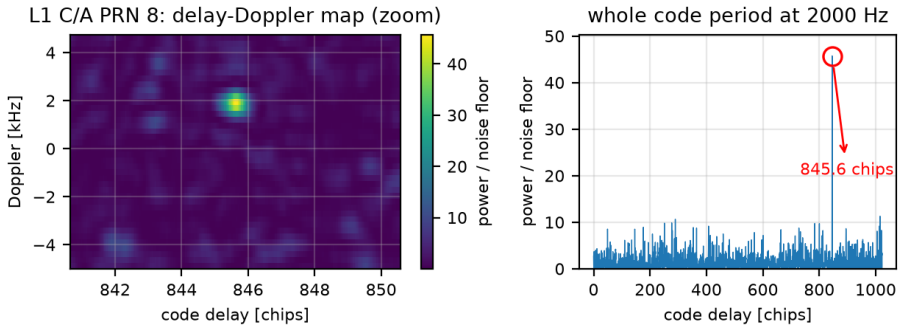


Figure 32: Acquisition, L1 C/A: delay–Doppler search of PRN 8; one sharp peak (test statistic 59 vs threshold 47) at 2000 Hz and 845.6 chips.

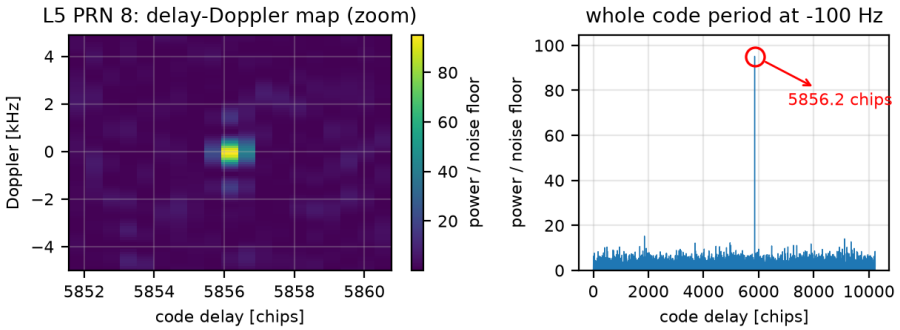


Figure 33: Acquisition, L5: delay–Doppler search of PRN 8; one sharp peak (test statistic 134 vs threshold 49) at 1392 Hz and 5856.2 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold[9]
L1 C/A	6	1	8	1.2
L5	2	2	8	2.7

Table 30: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

[9]Margin of the band's strongest detection: the largest acquisition test statistic among the detected attempts (the correlation peak of the delay–Doppler grid, normalised by the power of the input samples searched, i.e. the noise power, since GNSS signals are below the noise) divided by the detection threshold, which GNSS-SDR derives from the configured probability of false alarm. A value of 1 is right at the detection limit; e.g. 2.5 means the peak cleared the threshold 2.5 times over.

7.1.4 Tracking

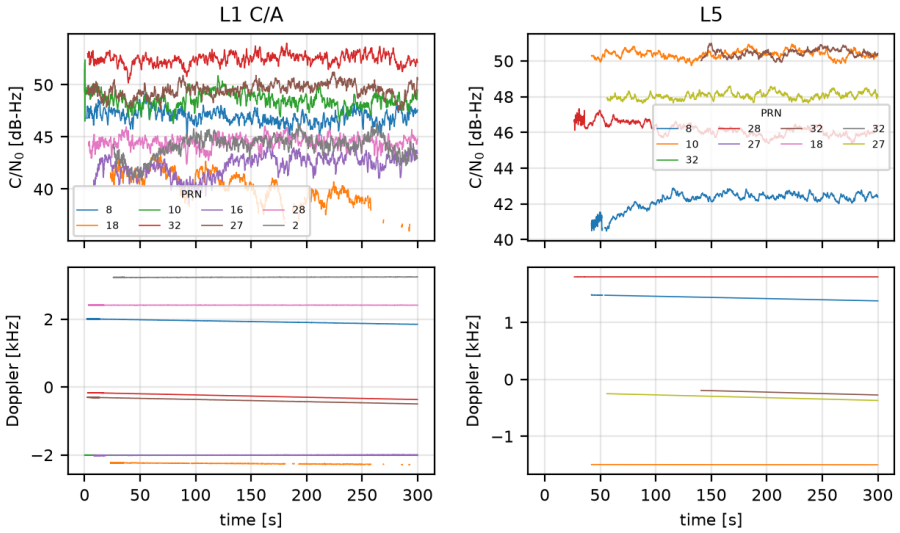


Figure 34: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

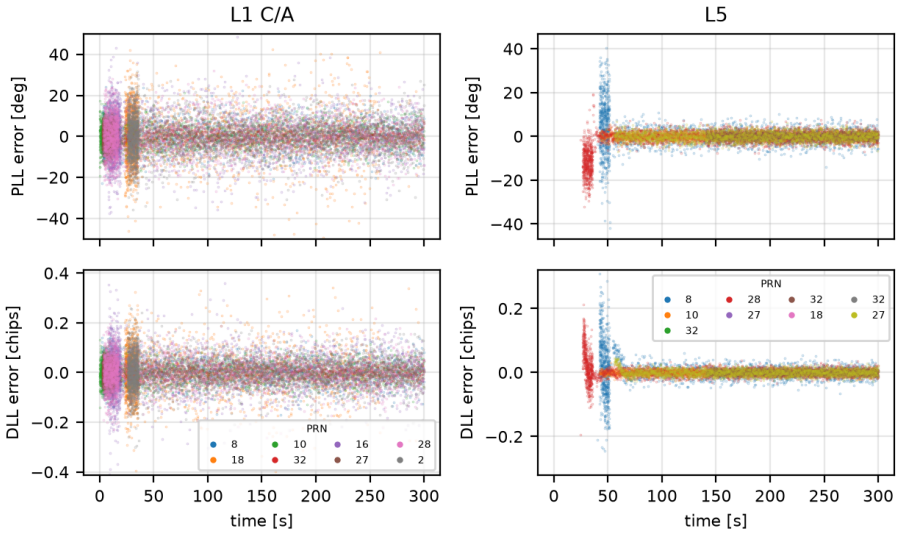


Figure 35: Tracking: loop errors per satellite while phase-locked, i.e. the discriminator outputs that the loops drive to zero: PLL carrier-phase error (degrees) and DLL code-delay error (chips) of every correlation. Noise centred on zero = clean tracking; its spread grows as C/N_0 drops, and jumps would mark cycle slips or a loop losing lock.

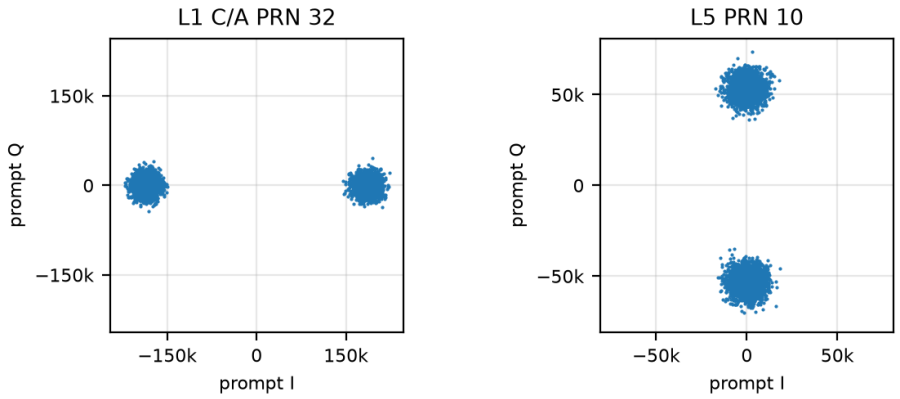


Figure 36: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or, for the dataless pilots tracked for Galileo (E1-C, E5a-Q), the ±1 chips of their secondary code, mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]	PLL error std [deg]	DLL error std [chips]
L1 C/A	0	8	47.2	99	0	298	6.3	0.048
L1 C/A	1	18	41.1	86	0	277	13.2	0.099
L1 C/A	2	10	49.1	100	0	300	5.2	0.038
L1 C/A	3	32	52.4	95	0	299	3.5	0.025
L1 C/A	4	16	42.3	95	0	293	11.1	0.085
L1 C/A	5	27	49.5	95	0	299	4.8	0.036
L1 C/A	6	28	44.4	91	0	299	8.9	0.063
L1 C/A	7	2	43.6	96	0	275	9.5	0.068
L5	8	8	41.2	66	1	276	7.8	0.052
L5	9	10	50.4	68	0	282	1.5	0.007
L5	10	32	29.1	0	2	32	–	–
L5	11	28	46.3	95	0	275	6.7	0.032
L5	13	27	28.9	0	3	16	–	–
L5	13	32	47.7	29	3	210	1.5	0.007
L5	14	18	28.6	0	3	16	–	–
L5	14	32	29.0	0	3	32	–	–
L5	15	27	48.0	68	0	260	1.9	0.010

Table 31: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8; loop error standard deviations while phase-locked).

7.1.5 Telemetry decoder

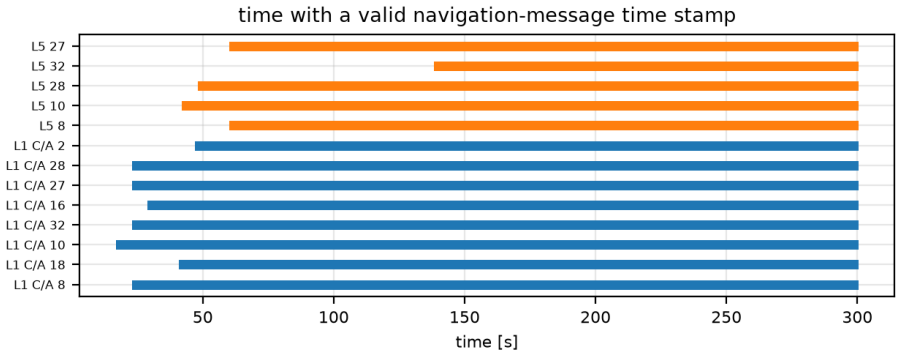


Figure 37: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
L1 C/A	2	43	yes	47.1	500472.0
L1 C/A	8	47	yes	23.1	500448.0
L1 C/A	10	48	yes	17.1	500442.0
L1 C/A	16	46	yes	29.1	500454.0
L1 C/A	18	44	yes	41.2	500466.0
L1 C/A	27	47	yes	23.1	500448.0
L1 C/A	28	47	yes	23.1	500448.0
L1 C/A	32	47	yes	23.1	500448.0
L5	8	25	yes	60.3	500485.1
L5	10	26	yes	42.1	500467.0
L5	27	25	yes	60.3	500485.2
L5	28	32	yes	48.4	500473.3
L5	32	16	yes	138.5	500563.3

Table 32: Telemetry decoder: L1 C/A LNAV subframes, L5 CNAV messages decoded, ephemeris obtained, and the first valid time of week.

7.1.6 Observables

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

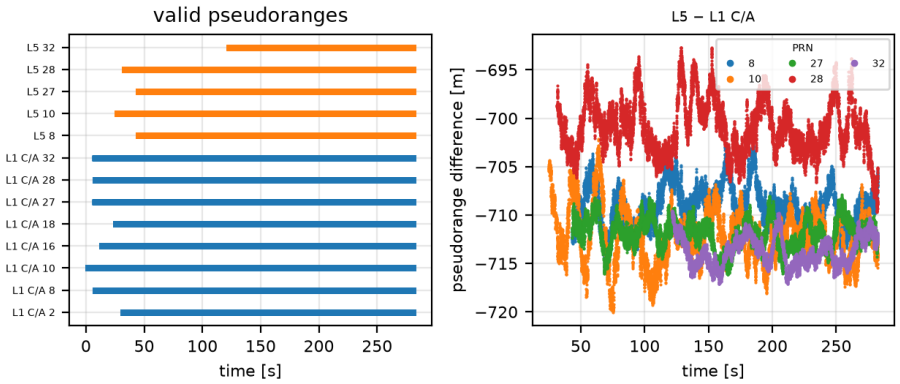


Figure 38: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
L5 − L1 C/A	8	11972	80	-709.04	2.07	-0.0001
L5 − L1 C/A	10	12837	86	-712.22	2.77	-0.0056
L5 − L1 C/A	27	11970	80	-711.92	1.52	-0.0025
L5 − L1 C/A	28	12563	84	-700.94	2.75	-0.0052
L5 − L1 C/A	32	8062	54	-713.71	1.50	0.0021

Table 33: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

7.1.7 PVT

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

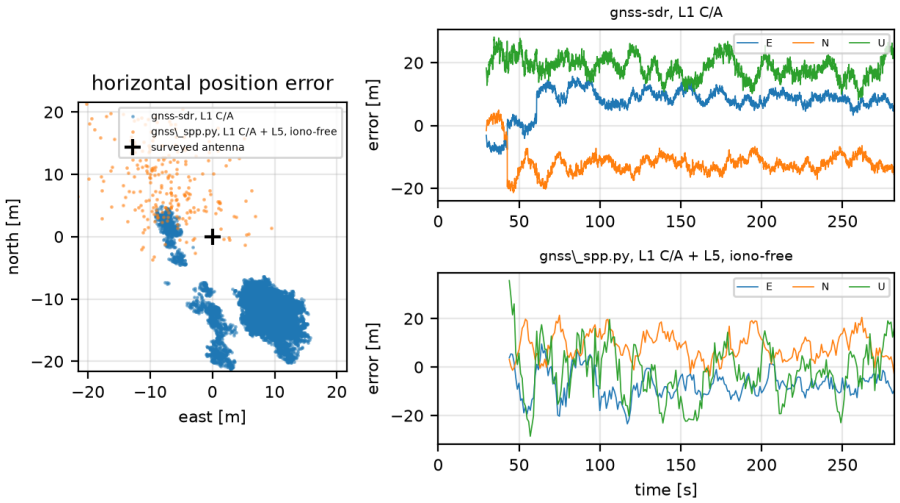


Figure 39: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution (single frequency, broadcast ionosphere model). gnss_spp.py: computed from gnss-sdr's observables and decoded GPS ephemeris, removing the ionospheric delay with the second band (pseudoranges carrier-smoothed over 100 s).

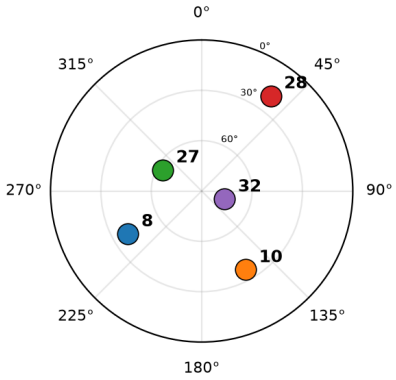


Figure 40: PVT: sky plot of the GPS satellites in the gnss_spp.py dual-frequency solution (azimuth, elevation) during the recording.

Solution	gnss-sdr, L1 C/A	gnss_spp.py, L1 C/A + L5, iono-free
Solutions	13133	239
Run with fix [%]	90	85
Mean E / N / U [m]	7.7 / -11.8 / 18.5	-7.3 / 8.5 / -2.1
Std E / N / U [m]	4.5 / 3.8 / 3.5	5.7 / 5.5 / 11.1
RMS 2-D [m]	15.3	13.8
RMS 3-D [m]	24.3	17.8
95% horiz. [m]	20.1	22.2

Table 34: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp.py computes one per second).

7.1.8 Signal conditioner (Galileo E1 + E5a)

Channel	ch0 (E1)	ch1 (E5a)
File format	ci8	ci8
Source item type	ibyte	ibyte
Data type adapter	Ibyte_To_Complex	Ibyte_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	×5 zero-order hold, offline (`upsample_iq.py`)	Pass_Through
Rate [MS/s]	4 → 20	20 → 20

Table 35: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

7.1.9 Acquisition (Galileo E1 + E5a)

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

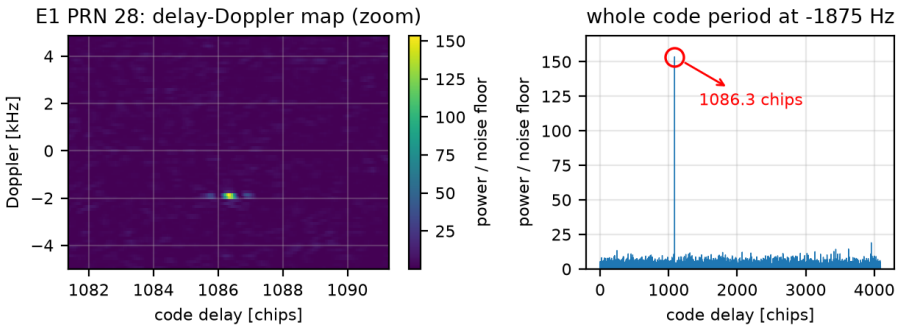


Figure 41: Acquisition, E1: delay–Doppler search of PRN 28; one sharp peak (test statistic 215 vs threshold 52) at -1875 Hz and 1086.3 chips.

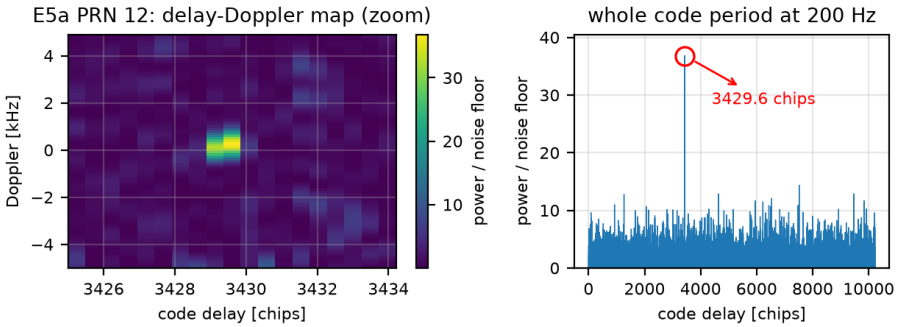


Figure 42: Acquisition, E5a: delay–Doppler search of PRN 12; one sharp peak (test statistic 51 vs threshold 49) at -492 Hz and 3429.6 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold[10]
E1	9	4	28	4.1
E5a	2	1	12	1.0

Table 36: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

[10]Margin of the band's strongest detection: the largest acquisition test statistic among the detected attempts (the correlation peak of the delay–Doppler grid, normalised by the power of the input samples searched, i.e. the noise power, since GNSS signals are below the noise) divided by the detection threshold, which GNSS-SDR derives from the configured probability of false alarm. A value of 1 is right at the detection limit; e.g. 2.5 means the peak cleared the threshold 2.5 times over.

7.1.10 Tracking (Galileo E1 + E5a)

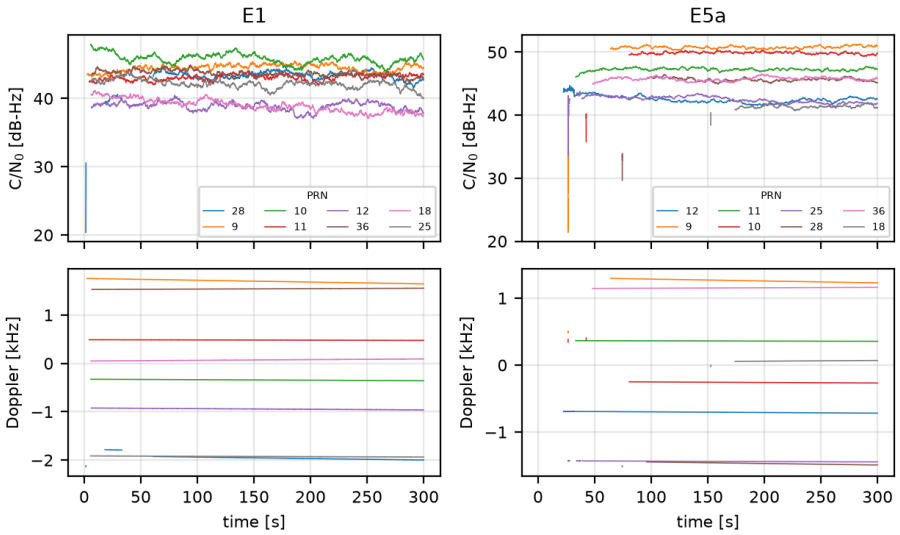


Figure 43: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

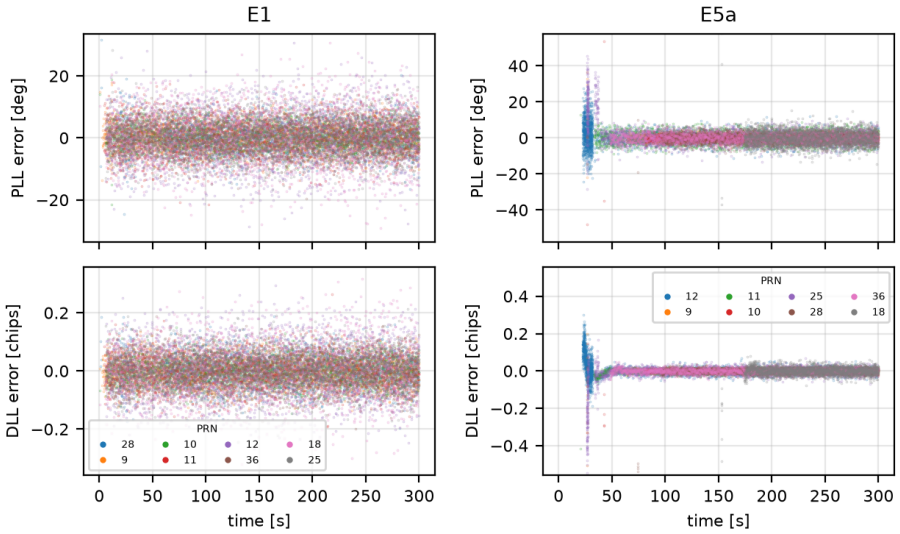


Figure 44: Tracking: loop errors per satellite while phase-locked, i.e. the discriminator outputs that the loops drive to zero: PLL carrier-phase error (degrees) and DLL code-delay error (chips) of every correlation. Noise centred on zero = clean tracking; its spread grows as C/N_0 drops, and jumps would mark cycle slips or a loop losing lock.

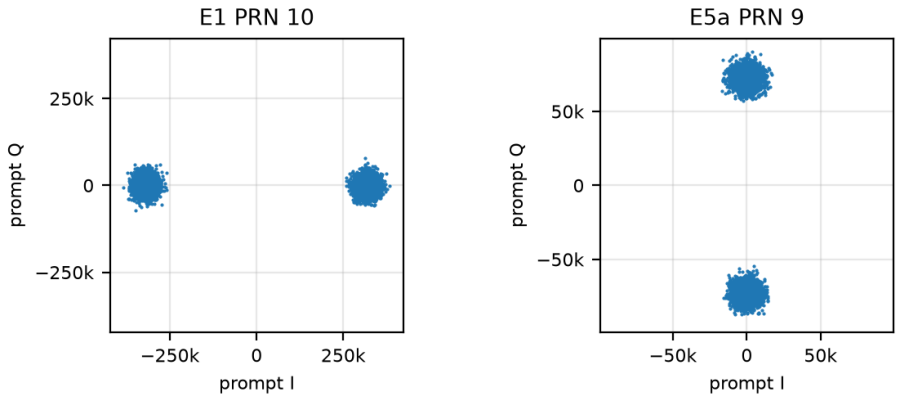


Figure 45: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or, for the dataless pilots tracked for Galileo (E1-C, E5a-Q), the ±1 chips of their secondary code, mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]	PLL error std [deg]	DLL error std [chips]
E1	0	28	43.4	85	3	299	5.2	0.052
E1	1	9	44.6	99	0	300	4.2	0.044
E1	2	10	45.9	98	0	300	3.9	0.041
E1	3	11	43.1	99	0	300	5.5	0.053
E1	4	12	39.0	98	0	300	8.2	0.088
E1	5	36	43.6	98	0	298	4.8	0.052
E1	6	18	39.0	98	0	299	8.4	0.089
E1	7	25	42.2	98	0	299	5.8	0.062
E5a	8	12	42.9	92	0	279	6.5	0.054
E5a	9	9	50.4	30	1	273	4.6	0.023
E5a	10	11	47.0	54	0	280	3.6	0.014
E5a	11	10	29.8	20	2	273	3.6	0.013
E5a	12	25	42.9	60	0	273	7.3	0.069
E5a	13	28	45.7	47	0	226	3.4	0.024
E5a	14	36	45.4	52	0	273	2.4	0.012
E5a	15	18	41.2	35	0	147	4.7	0.025

Table 37: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8; loop error standard deviations while phase-locked).

7.1.11 Telemetry decoder (Galileo E1 + E5a)

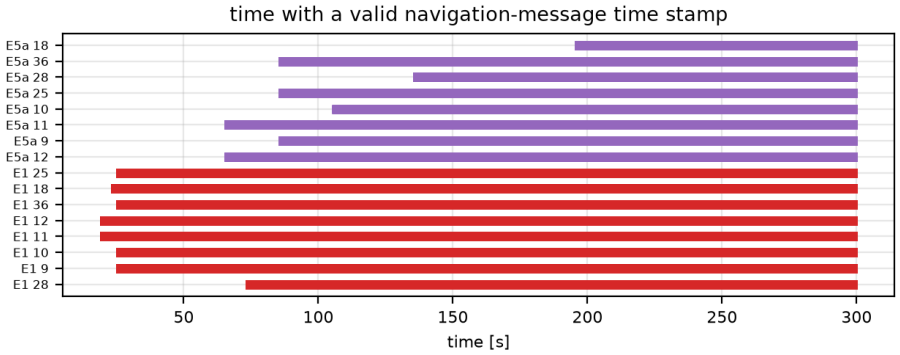


Figure 46: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
E1	9	26	yes	25.2	500450.0
E1	10	27	yes	25.2	500450.0
E1	11	27	yes	19.2	500444.0
E1	12	27	yes	19.2	500444.0
E1	18	26	yes	23.2	500448.0
E1	25	26	yes	25.2	500450.0
E1	28	22	yes	73.2	500498.0
E1	36	27	yes	25.2	500450.0
E5a	9	12	yes	85.4	500510.3
E5a	10	11	yes	105.4	500530.3
E5a	11	13	yes	65.4	500490.3
E5a	12	13	yes	65.4	500490.3
E5a	18	4	yes	195.4	500620.3
E5a	25	12	yes	85.4	500510.3
E5a	28	9	yes	135.4	500560.3
E5a	36	12	yes	85.4	500510.3

Table 38: Telemetry decoder: E1 I/NAV words, E5a F/NAV pages decoded, ephemeris obtained, and the first valid time of week.

7.1.12 Observables (Galileo E1 + E5a)

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

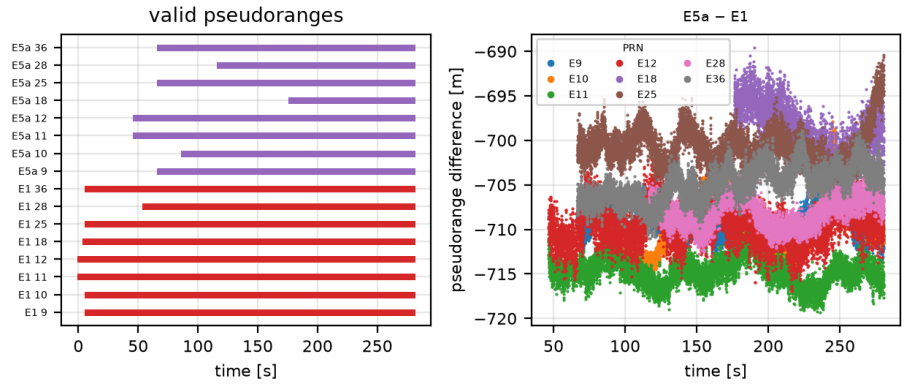


Figure 47: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
E5a − E1	9	10631	71	-709.38	1.88	-0.0007
E5a − E1	10	9645	64	-709.26	3.55	0.0208
E5a − E1	11	11505	77	-714.67	1.53	-0.0035
E5a − E1	12	11520	77	-709.41	2.18	0.0012
E5a − E1	18	5144	34	-698.88	2.95	-0.0589
E5a − E1	25	10637	71	-700.71	2.13	0.0007
E5a − E1	28	8150	54	-708.06	1.63	0.0079
E5a − E1	36	10632	71	-704.86	1.92	0.0213

Table 39: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

7.1.13 PVT (Galileo E1 + E5a)

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

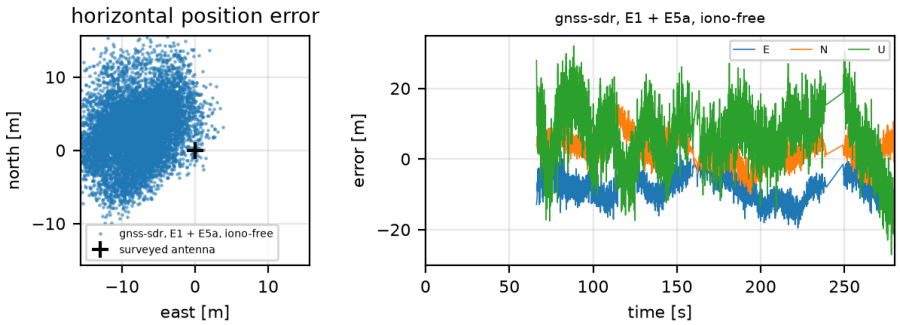


Figure 48: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution.

Solution	**gnss-sdr, E1 + E5a, iono-free**
Solutions	9278
Run with fix [%]	73
Mean E / N / U [m]	-8.1 / 2.7 / 6.0
Std E / N / U [m]	3.8 / 4.0 / 8.4
RMS 2-D [m]	10.2
RMS 3-D [m]	14.5
95% horiz. [m]	15.0

Table 40: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp.py computes one per second).

7.2 Second record

Parameter	Channel 0	Channel 1
Start time	2026-10-23 19:33:16 UTC	
Frequency	1575.420 MHz (L1)	1227.600 MHz (L2)
Duration	300.0 s	300.0 s
Sample rate	4.000 MS/s	4.000 MS/s
Sample type	signed complex 16-bit samples (I+Q), little-endian, as received from the X310 (sc16)	
Gain	25.0 dB	25.0 dB
Antenna	RX1	RX2
File size	4800.0 MB	4800.0 MB
Samples written	1,200,000,000	1,200,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	
GNSS-SDR configuration	`l1ca16bits4msps_l2c16bits4msps.conf`	

Table 41: Metadata for recording `x310_20261023T193310Z`, captured starting 2026-10-23T19:33:16.661684Z.

7.2.1 Signal source

Raw I/Q samples of each channel analysed within the window [10, 10.5] s after the start recording instant.

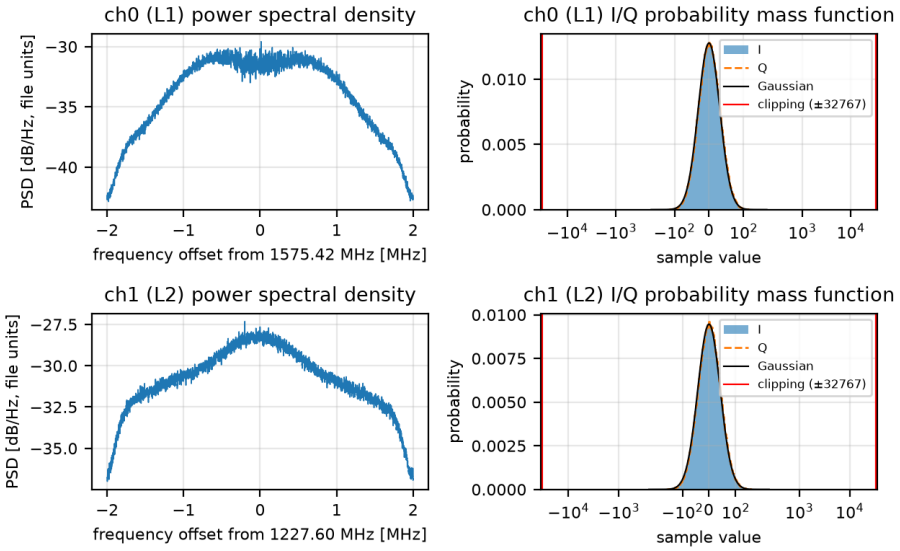


Figure 49: Signal source: power spectral density (the receiver passband over a flat thermal-noise floor; GNSS signals are below the noise, so only interference would stand out) and probability mass functions of the integer I/Q sample values (Gaussian noise, centred, not clipped).

Channel	**ch0**	**ch1**
Frequency [MHz]	1575.42	1227.60
Rate [MS/s]	4	4
Format	ci16_le	ci16_le
Bytes per complex sample	4 (2 for I, 2 for Q)	4 (2 for I, 2 for Q)
Full scale	±32767	±32767
Gain [dB]	25	25
Std I / Q	31.2 / 31.2	42.1 / 42.2
Mean I / Q	-0.04 / 0.06	0.03 / -0.02
Min / max I	-164 / 155	-205 / 216
Min / max Q	-155 / 165	-203 / 212
Headroom[11]	1048.7	777.0
Clipped [%]	0.000	0.000
Lost samples	0	0

Table 42: Signal source: sample statistics per channel (headroom: how many standard deviations of the noise fit before clipping).

[11] Headroom = A/σ, where A is the full scale of the sample format (127 for 8-bit, 32767 for 16-bit) and σ the standard deviation of the I and Q samples together (both in file units, over the analysed window). For Gaussian noise, the fraction of clipped values is $2Q(A/\sigma)$, with Q the Gaussian tail function: e.g. 0.003 % at 4.2σ.

7.2.2 Signal conditioner

Channel	ch0 (L1 C/A, E1)	ch1 (L2C)
File format	ci16_le	ci16_le
Source item type	ishort	ishort
Data type adapter	Ishort_To_Complex	Ishort_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	Pass_Through	Pass_Through
Rate [MS/s]	4 → 4	4 → 4

Table 43: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

7.2.3 Acquisition

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

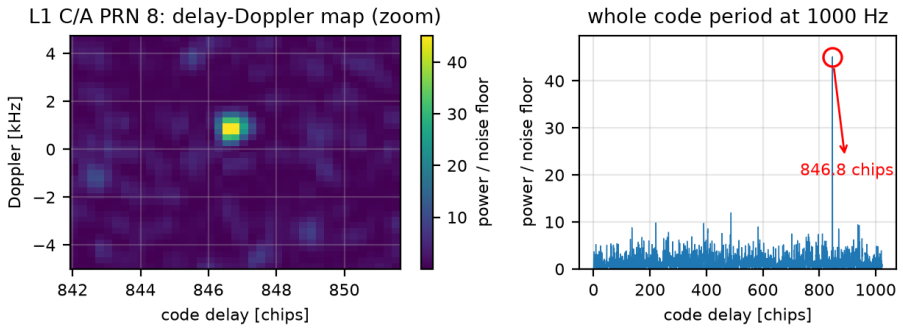


Figure 50: Acquisition, L1 C/A: delay–Doppler search of PRN 8; one sharp peak (test statistic 62 vs threshold 44) at 1000 Hz and 846.8 chips.

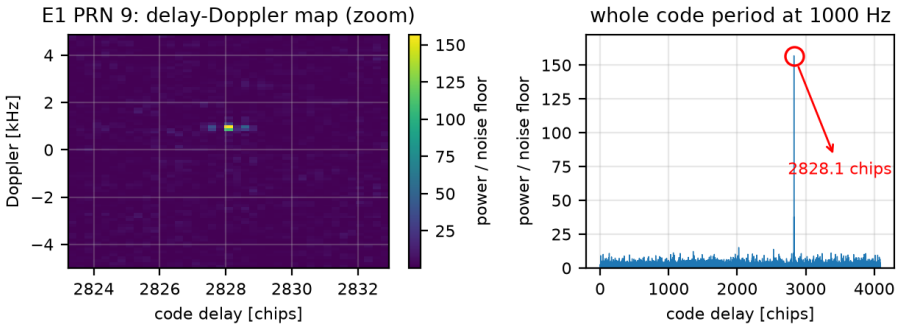


Figure 51: Acquisition, E1: delay–Doppler search of PRN 9; one sharp peak (test statistic 211 vs threshold 48) at 1000 Hz and 2828.1 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold[12]
L1 C/A	7	1	8	1.4
L2C	not dumped (search grid too large); acquired satellites: see the tracking table			
E1	1	1	9	4.4

Table 44: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or do not broadcast that signal, e.g. older GPS satellites on L5/L2C) are rejected below the threshold.

[12]Margin of the band's strongest detection: the largest acquisition test statistic among the detected attempts (the correlation peak of the delay–Doppler grid, normalised by the power of the input samples searched, i.e. the noise power, since GNSS signals are below the noise) divided by the detection threshold, which GNSS-SDR derives from the configured probability of false alarm. A value of 1 is right at the detection limit; e.g. 2.5 means the peak cleared the threshold 2.5 times over.

7.2.4 Tracking

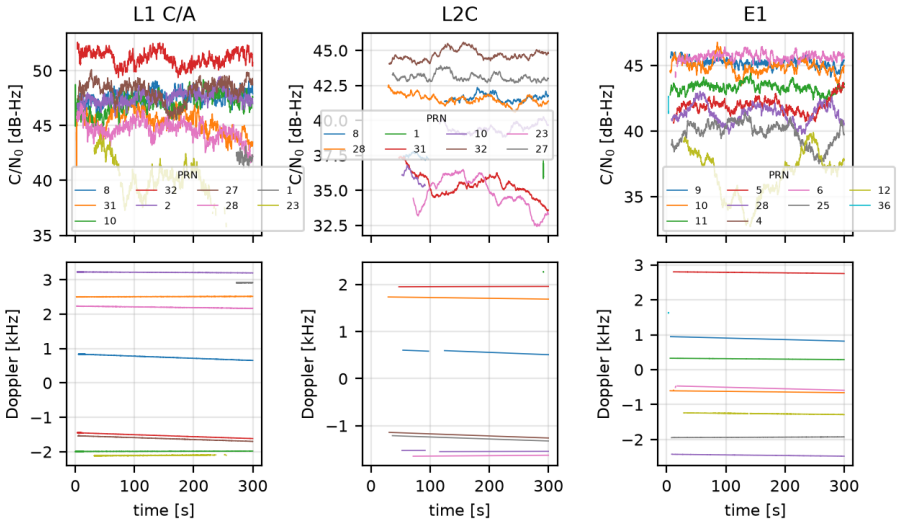


Figure 52: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

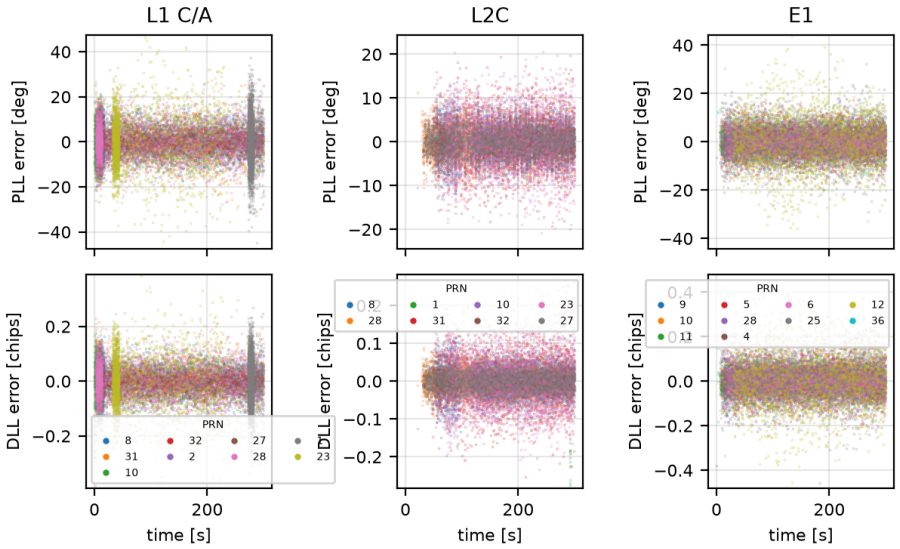


Figure 53: Tracking: loop errors per satellite while phase-locked, i.e. the discriminator outputs that the loops drive to zero: PLL carrier-phase error (degrees) and DLL code-delay error (chips) of every correlation. Noise centred on zero = clean tracking; its spread grows as C/N_0 drops, and jumps would mark cycle slips or a loop losing lock.

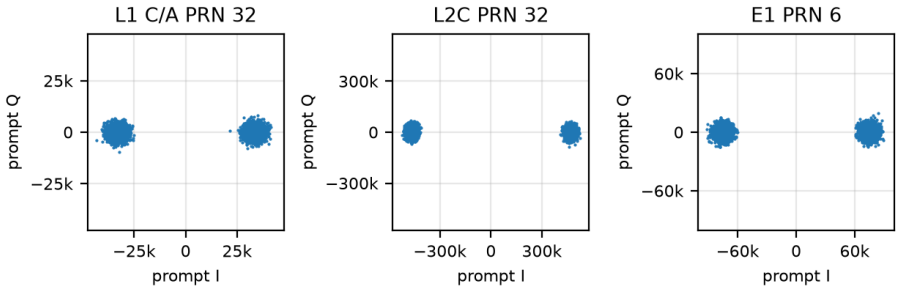


Figure 54: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis), or, for the dataless pilots tracked for Galileo (E1-C, E5a-Q), the ±1 chips of their secondary code, mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot \text{full scale} \cdot \sqrt{2}$ a sum of N samples can reach.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]	PLL error std [deg]	DLL error std [chips]
L1 C/A	0	8	47.3	91	0	297	6.1	0.049
L1 C/A	1	31	45.8	100	0	298	7.6	0.061
L1 C/A	2	10	46.7	100	0	300	6.7	0.053
L1 C/A	3	32	51.5	94	0	298	3.9	0.030
L1 C/A	4	2	47.3	95	0	298	6.1	0.047
L1 C/A	5	27	48.2	88	0	299	5.7	0.043
L1 C/A	6	28	45.3	96	0	299	8.1	0.061
L1 C/A	7	1	42.2	79	1	30	11.1	0.085
L1 C/A	7	23	42.7	83	1	232	12.3	0.095
L2C	8	8	41.4	80	1	274	3.5	0.035
L2C	9	28	41.5	97	0	279	3.3	0.031
L2C	10	1	36.7	9	0	9	52.8	0.059
L2C	11	31	35.4	91	0	280	5.9	0.065
L2C	12	10	39.3	80	1	280	4.4	0.045
L2C	13	32	44.6	97	0	279	2.7	0.022
L2C	14	23	34.8	92	0	249	6.1	0.070
L2C	15	27	43.1	95	0	279	2.9	0.026
E1	16	9	45.2	98	0	300	3.7	0.043
E1	17	10	44.9	98	0	300	4.3	0.048
E1	18	11	43.3	98	0	300	5.4	0.055
E1	19	5	41.9	97	0	297	5.8	0.068
E1	20	28	41.3	98	0	298	6.0	0.070
E1	21	4	26.4	1	1	11	–	–
E1	21	6	45.7	99	1	286	4.5	0.042
E1	22	25	40.0	98	0	298	7.5	0.086
E1	23	12	37.1	97	1	276	11.2	0.125
E1	23	36	44.3	0	1	21	48.0	0.107

Table 45: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8; loop error standard deviations while phase-locked).

7.2.5 Telemetry decoder

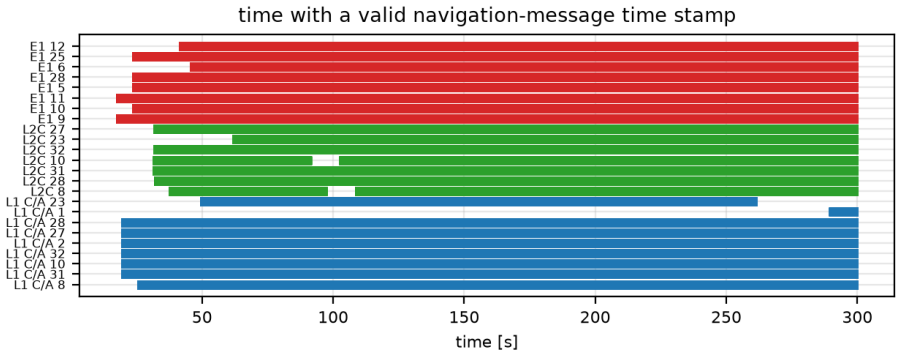


Figure 55: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
E1	5	27	yes	23.5	502438.0
E1	6	23	yes	45.5	502460.0
E1	9	26	yes	17.5	502432.0
E1	10	27	yes	23.5	502438.0
E1	11	27	yes	17.5	502432.0
E1	12	24	yes	41.5	502456.0
E1	25	27	yes	23.5	502438.0
E1	28	27	yes	23.5	502438.0
L1 C/A	1	2	no	289.4	502704.0
L1 C/A	2	46	yes	19.4	502434.0
L1 C/A	8	45	yes	25.4	502440.0
L1 C/A	10	45	yes	19.4	502434.0
L1 C/A	23	36	yes	49.4	502464.0
L1 C/A	27	43	yes	19.4	502434.0
L1 C/A	28	45	yes	19.4	502434.0
L1 C/A	31	47	yes	19.4	502434.0
L1 C/A	32	45	yes	19.4	502434.0
L2C	8	8	yes	37.4	0.0
L2C	10	8	yes	31.5	0.0
L2C	23	13	yes	61.6	0.0
L2C	27	12	yes	31.6	0.0
L2C	28	16	yes	31.8	0.0
L2C	31	12	yes	31.4	0.0
L2C	32	12	yes	31.5	0.0

Table 46: Telemetry decoder: L1 C/A LNAV subframes, L2C CNAV messages, E1 I/NAV words decoded, ephemeris obtained, and the first valid time of week.

7.2.6 Observables

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

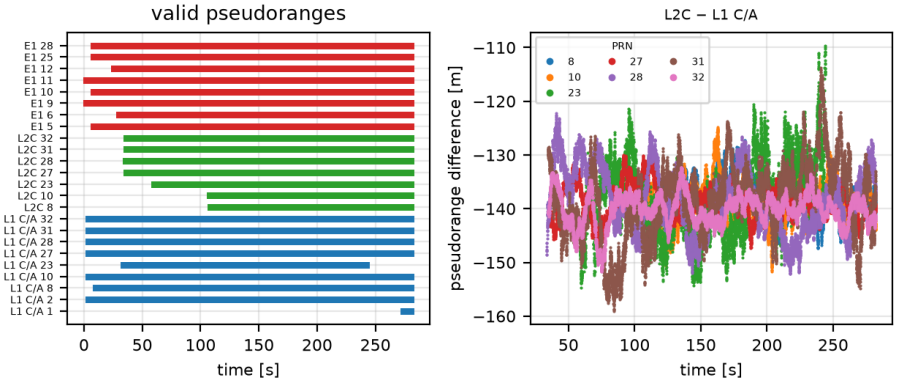


Figure 56: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

Difference	PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
L2C – L1 C/A	8	8789	59	-137.54	3.60	-0.0130
L2C – L1 C/A	10	8829	59	-138.91	3.87	0.0045
L2C – L1 C/A	23	9286	62	-138.17	6.94	0.0399
L2C – L1 C/A	27	12381	83	-138.90	3.03	-0.0087
L2C – L1 C/A	28	12437	83	-138.47	5.77	-0.0274
L2C – L1 C/A	31	12393	83	-138.52	7.00	0.0405
L2C – L1 C/A	32	12387	83	-140.16	2.89	0.0019

Table 47: Observables: pseudorange difference between the two bands of each satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

7.2.7 PVT

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

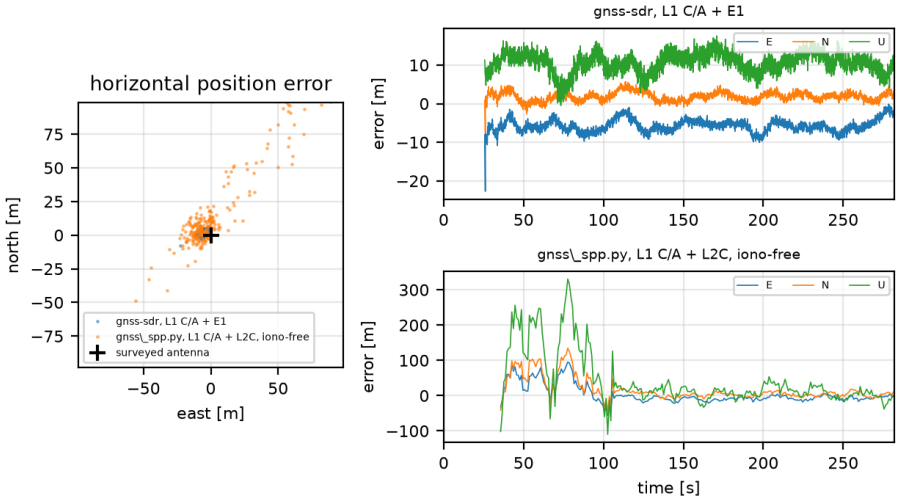


Figure 57: PVT: position error with respect to the surveyed antenna position. gnss-sdr: GNSS-SDR's own RTKLIB solution (single frequency, broadcast ionosphere model). gnss_spp.py: computed from gnss-sdr's observables and decoded GPS ephemeris, removing the ionospheric delay with the second band (pseudoranges carrier-smoothed over 100 s).

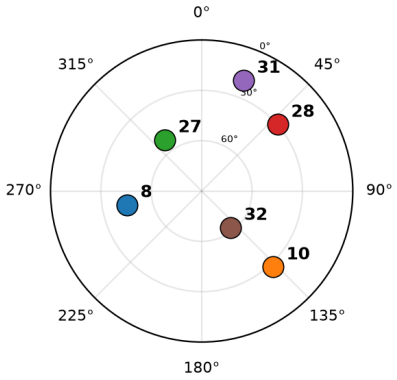


Figure 58: PVT: sky plot of the GPS satellites in the gnss_spp.py dual-frequency solution (azimuth, elevation) during the recording.

Solution	**gnss-sdr, L1 C/A + E1**	**gnss_spp.py, L1 C/A + L2C, iono-free**
Solutions	13328	247
Run with fix [%]	91	88
Mean E / N / U [m]	-5.8 / 1.8 / 10.6	3.3 / 16.8 / 42.9
Std E / N / U [m]	1.6 / 1.2 / 2.5	26.5 / 32.0 / 78.1
RMS 2-D [m]	6.4	44.9
RMS 3-D [m]	12.6	99.8
95% horiz. [m]	8.4	116.1

Table 48: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, gnss_spp.py computes one per second).